

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA

hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: PhD. Trịnh Văn Chiến

Department: Faculty of Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

Student's engagement detection in online classes

PHAN MINH HÒA

hoa.pm225495@sis.hust.edu.vn

Program: Data Science and Artificial Intelligence

Supervisor: Dr. Trịnh Văn Chiến

Signature

Department: Faculty of Computer Science

School: School of Information and Communications Technology

HANOI, 06/2026

ACKNOWLEDGMENT

I am grateful to my supervisor, **PhD. Trịnh Văn Chiến**, for the guidance, careful feedback, and patience that shaped this thesis. I thank the lecturers and staff of the School of Information and Communications Technology at Hanoi University of Science and Technology for the foundation they gave me, and the creators of the CMOSE [1] and DAiSEE [2] datasets, whose work made this study possible. Finally, I thank my family and friends for their constant support, and acknowledge my own persistence in seeing this work through.

ABSTRACT

Recognising how engaged a student is from webcam video can help teachers in online and hybrid classes, but it is difficult: the engagement labels are ordinal (Highly-Disengage, Disengage, Engage, Highly-Engage), heavily imbalanced, and partly subjective, and most studies do not have reliable evaluation for generalisation. Existing methods share two further weaknesses: they feed all of the facial features into one model as a single block, and they judge results mainly by accuracy, which stays high even when a model ignores the rare disengaged classes. In this thesis I address both problems. Because the facial features measure different behaviours that change at different speeds, I split the 709 OpenFace features into five groups (gaze, eye landmarks, face landmarks, head pose, and action units) and give each group its own temporal encoder, chosen from a Temporal Convolutional Network, an LSTM, or a Transformer; each group can also be combined with an I3D motion stream from the whole clip, and every stream has a small extra training head. Because accuracy is misleading on imbalanced ordinal labels, I select models with quadratic-weighted kappa (QWK) and report macro-accuracy and macro-MAE only as secondary checks. I make three contributions. First, I propose this feature-group hybrid and test all 243 ways of assigning encoders to groups, with and without I3D; on CMOSE the best hybrid reaches a QWK of 0.605 against 0.537 for the best single-encoder baseline, and only head pose clearly prefers one encoder, a TCN. Second, I collect and hand-label a private webcam test set, built with the same feature pipeline and kept only for testing. Third, I run a 3×3 cross-dataset study that trains on CMOSE, DAiSEE, or both and tests on each public set and the private set. Models barely transfer: off-diagonal QWK falls to near zero even when accuracy still looks acceptable. On the unseen private set, training on both datasets with the hybrid gives the best result (QWK 0.379 against 0.285 for the best baseline). I conclude that quadratic-weighted kappa, not accuracy, must select engagement models when the labels are imbalanced.

Keywords: student engagement recognition, affective computing, OpenFace, I3D, temporal convolutional networks, ordinal classification, class imbalance, cross-dataset generalisation.

Student
(Signature and full name)

TABLE OF CONTENTS

CHAPTER 1. INTRODUCTION.....	1
1.1 Problem Statement.....	1
1.2 Background and Problems of Research	3
1.3 Research Objectives and Conceptual Framework	4
1.4 Contributions	5
1.5 Organization of Thesis	6
CHAPTER 2. LITERATURE REVIEW	8
2.1 Scope of Research	8
2.2 Related Work	8
2.3 Feature Extraction	10
2.3.1 OpenFace 2.0 facial-behaviour descriptors	10
2.3.2 I3D motion and appearance features	10
2.4 Temporal Architectures	10
2.4.1 Temporal Convolutional Networks (TCN)	11
2.4.2 Long Short-Term Memory (LSTM)	11
2.4.3 Transformers	12
2.5 Loss Functions for Imbalance and Ordinality	13
2.6 Evaluation Metrics.....	14
2.7 Statistical Measures for Result Analysis	16
2.8 Chapter Summary.....	17
CHAPTER 3. METHODOLOGY.....	18
3.1 Overview	18
3.2 Data and Preprocessing	18
3.2.1 Datasets	18

3.2.2 The private evaluation set	20
3.2.3 Features and preprocessing	21
3.3 Baseline Models	21
3.4 Grouping the Facial Features by Meaning	24
3.5 Hybrid Architecture (Proposed)	24
3.6 Loss Functions	26
3.7 Training Protocol.....	27
3.8 Evaluation Protocol	27
3.9 Chapter Summary.....	28
CHAPTER 4. BASELINE MODELS.....	29
4.1 Overview	29
4.2 Evaluation Protocol	29
4.2.1 Metrics	29
4.2.2 Development Corpus	30
4.2.3 Development Loss.....	31
4.3 In-Domain Results	32
4.4 Prediction Agreement	33
4.4.1 Agreement between Configurations	33
4.4.2 OpenFace versus I3D.....	34
4.5 Cross-Dataset Generalisation	35
4.5.1 The 3×3 Matrix	35
4.5.2 Closing the Loop: Metric and Loss Reliability	37
4.5.3 In-Domain versus Transfer.....	37
4.6 Chapter Summary.....	37
CHAPTER 5. FEATURE-GROUP HYBRID	39
5.1 Overview	39

5.2 Consistency with the Baseline Frame	39
5.3 Encoder Choice per Group	40
5.4 Effect of I3D Fusion	41
5.5 Comparison with Baselines	42
5.6 Cross-Dataset Generalisation	43
5.6.1 Cross-Dataset Matrix.....	43
5.6.2 The Private Set	43
5.6.3 In-Domain versus Transfer	46
5.7 Discussion	47
5.8 Chapter Summary.....	48
CHAPTER 6. CONCLUSIONS	49
6.1 Summary	49
6.2 Synthesis	50
6.3 Revisiting the Research Questions	51
6.4 Limitations.....	52
6.5 Practical Recommendations	53
6.6 Suggestion for Future Works.....	53
6.7 Closing Remarks	54
REFERENCE	56

LIST OF FIGURES

Figure 2.1	The residual TemporalBlock used in every TCN of this thesis: two weight-normalised dilated causal 1-D convolutions, each followed by a ReLU and dropout, summed with a residual connection (kernel size 3).	11
Figure 2.2	A single LSTM cell at time step t : the forget, input, and output gates regulate how the cell state C_t and hidden state h_t are updated from the previous states C_{t-1} , h_{t-1} and the input x_t	12
Figure 2.3	A single Transformer encoder layer used in this thesis: multi-head self-attention over the sinusoidally position-encoded input, followed by a position-wise feed-forward network, each sub-layer wrapped in a residual connection and layer normalisation.	13
Figure 3.1	Engagement class distribution by dataset.	19
Figure 3.2	The five single-encoder baselines. Each applies a single encoder to the full feature sequence (or pooled I3D vector) before classification.	23
Figure 3.3	Feature-group hybrid architecture. Each OpenFace group is encoded independently into a 64-d embedding (group encoders are chosen per group from TCN / Transformer (T) / LSTM); the I3D clip vector is encoded by an MLP into a 128-d embedding. The six embeddings are concatenated (448-d) and classified by a shared MLP; each stream also feeds an auxiliary head. The OpenFace-only variant drops the I3D row and fuses a 320-d vector.	25
Figure 3.4	The three interchangeable per-group encoders. Each maps a group's $T \times D_{\text{group}}$ sequence to a 64-dimensional embedding through temporal mean-pooling and layer normalisation; one is chosen independently for each of the five groups.	26
Figure 4.1	Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive τ means agreement). The metrics form a micro block and a macro block that barely agree; QWK is the only metric correlated with both.	31

Figure 4.2	What each loss trades: QWK (the selection metric) against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE. Macro-accuracy and macro-MAE improve as the loss is rebalanced, while QWK is generally highest under cross-entropy.	32
Figure 4.3	The five base models under the three losses on all six metrics, in-domain on CMOSE. The metrics disagree on the winner: accuracy, MAE, and Cohen’s kappa favour the I3D MLP under cross-entropy, QWK favours the OpenFace TCN, and the macro metrics peak under the balanced losses.	33
Figure 4.4	Pairwise prediction agreement (Cohen’s κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks. The diagonal blocks are the same architecture under different losses; agreement is moderate at best.	34
Figure 4.5	Cross-domain generalisation, best base model per cell (selected by QWK), on QWK (the selection metric) and the balanced secondary metrics macro-accuracy and macro-MAE, plus raw accuracy, the misleading one: accuracy stays high in exactly the off-diagonal cells where every chance-corrected metric collapses. .	36
Figure 5.1	Hybrid ablation across all 243 per-group encoder configurations, with and without I3D, on all six metrics (in-domain CMOSE). The dashed line in every panel is the same model (the single QWK-selected baseline: OpenFace TCN, cross-entropy) read on that panel’s metric, not a per-metric champion.	40
Figure 5.2	Cross-domain generalisation, best hybrid per cell (selected by QWK), on QWK (the selection metric) and the balanced secondary metrics macro-accuracy and macro-MAE, plus raw accuracy, which again stays high in the off-diagonal cells where the chance-corrected metrics collapse.	44
Figure 5.3	Cell-by-cell hybrid advantage: best-hybrid minus best-base QWK over the 3×3 train $\times$ test matrix (Figure 5.2 minus Figure 4.5). Every cell is positive (the hybrid is strongest across the whole matrix), and the gain is largest on the unseen, self-collected Private column. The in-domain CMOSE cell and the Private column (bold) are the two comparisons read in Sections 5.5 and 5.6.2.	45
Figure 5.4	Private-set confusion of the combined-trained best base model versus best hybrid (row-normalised).	46

Figure 5.5 Per-class F1 on the private set: combined-trained best base model versus best hybrid. The hybrid’s out-of-domain gain concentrates on the engaged region and the middle Disengage class, while Highly-Disengage collapses to zero. 46

Figure 5.6 In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out private set); each configuration is one point pairing its own two runs, so the axes pool no training populations. The correlation is weak for both the 486 hybrid configurations ($\rho = 0.27$) and the 15 baselines ($\rho = 0.31$). The hybrid cloud lies above and to the right of the baselines: the architecture improves both in-domain QWK and private-set transfer. 47

LIST OF TABLES

Table 3.1	Dataset statistics and class balance (%).	19
Table 3.2	Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning.	24
Table 4.1	The winning base configuration according to each metric (in-domain CMOSE).	30
Table 4.2	Best in-domain result per dataset (CMOSE vs DAiSEE).	31
Table 4.3	Best OpenFace encoder (selected on QWK) vs the I3D MLP on QWK and the balanced secondary metrics (in-domain CMOSE, cross-entropy).	35
Table 5.1	Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.	41
Table 5.2	Paired effect of adding the I3D stream on QWK, by evaluation regime (each OpenFace-only hybrid against its exact +I3D twin).	42
Table 5.3	Top-5 hybrid configs in-domain (CMOSE), sorted by QWK.	42
Table 5.4	Private set (test-only): best base vs best hybrid by training source.	44

LIST OF ABBREVIATIONS

Abbreviation	Definition
AU	Facial Action Unit
BiLSTM	Bidirectional Long Short-Term Memory network
CDF	Cumulative Distribution Function
CE	Cross-Entropy loss
CMOSE	Class-level Multimodal Online Student Engagement dataset
DAiSEE	Dataset for Affective States in E-Environments
EMD	Earth Mover’s Distance; here the squared-EMD / CDF-MSE ordinal loss
GELU	Gaussian Error Linear Unit
I3D	Inflated 3D ConvNet (spatiotemporal video feature extractor)
LSTM	Long Short-Term Memory network
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
QWK	Quadratic-Weighted Kappa
ReLU	Rectified Linear Unit
SMOTE	Synthetic Minority Over-sampling Technique
TCN	Temporal Convolutional Network

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Online and hybrid education has become a permanent part of modern teaching. Lectures, tutorials, and exams are now usually given through video calls and learning platforms, and much learning happens with a webcam as the only link between the student and the teacher. This change brings both an opportunity and a problem. The webcam stream reveals a great deal about how a student is following the material, but unlike in a physical classroom, the teacher cannot perceive the overall state of the class. A teacher facing dozens of simultaneous video tiles has no reliable way of noticing that one student lost focus several minutes earlier. Automatic engagement recognition aims to close this gap: it turns a short video of one student into an estimate of how engaged that student is, so that teachers, tutoring systems, and learning platforms can step in, change the pace of a lesson, or tell whether students are following along, on a scale that is unrealistic for teachers to do manually.

This thesis studies the exact form of that problem used by the research community and by the datasets I build on. The input is a short video clip of one student, about ten seconds long, recorded from a front webcam. The output is one of four ordinal levels: Highly-Disengage (HD), Disengage (DE), Engage (EG), and Highly-Engage (HE). These four levels form a scale from full withdrawal to active, focused participation, so the task is ordinal classification over four classes, not an unordered, nominal one.

Although it looks like ordinary video classification, engagement recognition is harder than a typical image- or action-recognition task for three reasons that shape the design choices in this thesis.

- (i) **The labels are ordinal, not nominal.** The four classes lie on a line: $HD < DE < EG < HE$. Predicting Highly-Engage for a Highly-Disengage student is a serious error, while predicting Engage for a Highly-Engage student is almost right. Plain accuracy ignores this order, treating both mistakes as equally wrong. So accuracy is the wrong training goal and the wrong evaluation metric. The size of an error must be weighted by how far apart the true and predicted classes are on the scale.
- (ii) **The labels are severely imbalanced.** Real classes are mostly attentive, so engagement datasets are dominated by the middle and upper levels. In the CMOSE dataset about 69% of clips are labelled Engage and fewer than 3% are Highly-Disengage (Section 3.2, Table 3.1). So a model can reach high accuracy

by always guessing the majority class and never predicting disengagement at all, which is the rare but important case that a monitoring system needs to catch.

- (iii) **The camera angle is never fixed.** Engagement is read from facial behaviour, but the camera geometry behind that behaviour changes from learner to learner: a laptop camera looks up from the desk, a built-in monitor camera sits near eye level, a phone is propped off to the side, so the same expression reaches the model at a different viewpoint, framing, and head pose every time. This matters because the OpenFace gaze, head-pose, and landmark features are all measured in the image geometry, so a change of camera angle moves the features even when the underlying engagement is identical. Each dataset is recorded with its own narrow range of setups, so a model trained on one dataset meets unfamiliar viewpoints the moment it is deployed. How robust models are to this change of angle is rarely measured, even though deployment always means cameras and framings the model has never seen.

Together, these three properties make the problem both practically important and technically interesting. A system that is accurate in aggregate but insensitive to disengagement, or that performs well on its training dataset but degrades sharply on unseen video, has little practical value in a classroom. So the main concern of this thesis is not overall accuracy but ordinal agreement on imbalanced data, and how well it holds when the data shifts.

These properties are not abstract: they follow directly from how an engagement signal would actually be used. The real unit of decision is rarely a single clip but a summary, such as an engagement curve over one session for one student, or a class-level spread the teacher watches in real time. The value of such a signal lies almost entirely in the rare cases: confirming that an already-attentive class is attentive provides the teacher with little information, while flagging the few moments when a student becomes disengaged is the main goal, the rare case that imbalance hides and that accuracy does not reward. A deployment also never sees the data it was tuned on: cameras, lighting, who is in the class, and the labelling rubric all change. So an in-domain number means little unless it is shown to hold under that change. This is why the thesis treats a self-collected, never-trained-on test set and a clear cross-dataset protocol as the setting in which any claim of usefulness must finally be judged.

1.2 Background and Problems of Research

Research on vision-based engagement recognition has, broadly, taken two routes. The first encodes each video frame with a facial-behaviour toolkit (most often OpenFace [3], which estimates gaze direction, head pose, facial landmarks, and action-unit intensities) and feeds the feature sequence to a classical classifier or a recurrent network, for example an OpenFace+BiLSTM pipeline [4] or classical models such as logistic regression, support vector machines, and gradient boosting on summary features [5]. The second route learns straight from pixels with deep video models, for instance an EfficientNetV2 backbone followed by an LSTM [6], or borrows context- and relationship-modelling ideas from affect recognition [7]. This thesis builds on two large datasets: CMOSE [1] is a recent, large, expert-labelled multimodal dataset and the only one of my sources that comes with ready OpenFace and I3D features, and DAiSEE [2] is a widely used public dataset recorded “in the wild” that gives only raw video, from which I make the same features myself. A systematic review of computer-vision engagement detection [8] lists these pipelines and confirms that class imbalance and subjective labels are common, unsolved problems across the field.

Two specific limits of the existing work motivate this thesis.

First, facial-behaviour features are mixed but are encoded as one block. A single OpenFace frame in CMOSE is a 709-dimensional vector that bundles together signals of very different physical kinds: a few gaze direction values, hundreds of eye- and face-landmark coordinates, head-pose shifts and rotations, and action-unit strengths. These parts move at different speeds: the eyes flick fast and noisily, head turns are slow and smooth, and action units fire in short bursts. The usual practice of joining all 709 numbers and pushing them through one sequence encoder forces a single temporal model onto signals that do not share a single timescale. It also makes the model harder to read, because the role of each behavioural channel is no longer separate.

Second, studies mostly use accuracy to evaluate, and do not have reliable evaluation for generalisation. Almost every engagement study reports plain accuracy as its headline number, and almost always in-domain, meaning train and test on the same dataset. Yet the only deployment that matters is on data the model has never seen, recorded with different cameras, viewpoints, and lighting than the training dataset. Because engagement labels are subjective and the class balance differs sharply between datasets (CMOSE is Engage-dominated while DAiSEE is split between Engage and Highly-Engage), it is reasonable to expect that models

transfer poorly, but without measuring it across datasets this remains an assumption. A further problem: on imbalanced data a model that simply predicts the majority class can reach a decent accuracy while having learned nothing that transfers, so the metric most papers report can hide the failure it is meant to reveal.

These two limits define the gap this thesis addresses. The mixed-feature problem calls for a representation that keeps the structure of the facial features instead of flattening it; the generalisation problem calls for a protocol that measures transfer directly, with metrics that cannot be tricked by predicting only the majority class.

1.3 Research Objectives and Conceptual Framework

The goal of this thesis is to improve, and to describe accurately, how engagement is recognised on a four-level ordinal scale from facial-behaviour and motion features. I do this through one architecture idea and one set of evaluation rules, written as four research questions.

- (i) **RQ1 — Decomposition.** Does splitting the OpenFace features into groups, each with its own temporal encoder, improve engagement classification over single-encoder baselines that encode all 709 features together?
- (ii) **RQ2 — Encoder assignment.** If splitting helps, which group→encoder choices actually matter? Is there a behavioural channel whose timing prefers one encoder family (convolutional, recurrent, or attention-based) over another?
- (iii) **RQ3 — Motion fusion.** Does adding a global I3D motion stream to the split model help, and does any gain hold across many configurations rather than arise from a single favourable one?
- (iv) **RQ4 — Generalisation and losses.** How well do these models generalise across datasets, and how does the loss trade ordinal agreement against recall on the minority classes and balanced ordinal error?

Conceptual framework. The shared idea behind RQ1–RQ3, shown by the hybrid architecture in Chapter 3 (Figure 3.3), is to treat a clip as several streams in time rather than one sequence. The OpenFace features are split into five meaningful groups (gaze, eye landmarks, face landmarks, head pose, and action units), and each group is encoded on its own into an embedding of fixed length by an encoder chosen to suit its timing. A global I3D clip embedding can be added as a sixth stream. The embeddings are joined and classified, and every stream also gets its own small head so that each behavioural channel has to be useful on its own. I call this “divide the face, then fuse” design the feature-group hybrid, and it is the main object tested in this thesis against single-encoder baselines, across datasets, and across loss

functions. The rule behind RQ4 is to select and rank every model on one metric, ordinal agreement (quadratic-weighted kappa, QWK), reporting balanced accuracy and balanced ordinal error (macro-MAE) only as secondary checks and never using plain accuracy to choose a model. Every model is tested not only in-domain but on two unseen datasets, one of which I collected myself.

1.4 Contributions

This thesis makes three contributions.

- (i) **Architecture — a feature-group hybrid that divides the face, then fuses.**
I propose a temporal architecture that splits the 709 OpenFace features into five behavioural groups, encodes each group with its own chosen Temporal Convolutional Network (TCN), Transformer, or LSTM, optionally adds a global motion stream from I3D, and trains every stream with a small head (Section 3.5). Trying all 243 ways to assign encoders to groups, with and without the I3D stream, shows that the I3D hybrid family scores higher than the strongest baseline (median QWK 0.553 against 0.537, best 0.605), that the gain belongs to the family rather than to a single favourable configuration, and that head pose is the only group with a clear encoder preference, preferring a TCN, a preference that also holds when the data shifts (Section 5.2).
- (ii) **Dataset — a new, self-collected private test set.** I collect 366 webcam clips, cut to a fixed length, filtered for tracking quality, labelled by hand by me through a purpose-built interface, and made with the same OpenFace/I3D pipeline as the public datasets. It is used only to test, never in training, how well a model works on real students recorded on their own webcams, where camera angles and conditions are unlike the curated public datasets (Section 3.2).
- (iii) **Generalisation — a cross-dataset study measured by QWK, not accuracy.** I run a 3×3 cross-dataset study (training on CMOSE, DAiSEE, or both together; testing on each public test set plus the private set) showing that engagement models do not transfer to a new dataset, that accuracy hides this collapse while QWK shows it, and that training on the combined datasets is the simplest fix that works best. On the private set the architecture and the combined-training strategy reinforce each other: the hybrid trained on both datasets is the best model (QWK 0.379 against 0.285 for the best baseline, a bigger gap than in-domain), which shows the practical value of all three contributions together (Sections 4.5 and 5.6).

These contributions are backed by a reproducible pipeline that turns about 1,500 trained runs into the figures and tables of this thesis, from feature extraction through training, cross-dataset evaluation, and figure/table generation.

1.5 Organization of Thesis

The remainder of this thesis is organised as follows.

Chapter 2 reviews the work on vision-based engagement recognition and introduces the building blocks the thesis relies on. It sets the scope of the study, looks at the related feature-based and end-to-end approaches with their strengths and weaknesses, and then presents the technical basics: the OpenFace and I3D feature extractors, the three families of temporal architecture (TCN, LSTM, Transformer), the loss functions used for imbalanced ordinal labels, and the ordinal metrics.

Chapter 3 presents the methodology. It gives an overview of the full pipeline as a single diagram, describes the three training datasets and their class imbalance, the private test set, and the preprocessing that makes them comparable, sets out the single-encoder baselines, and then gives the feature-group decomposition and the proposed hybrid built on top of it. It also establishes the loss functions and their forms, the training protocol, and the 3×3 cross-dataset protocol with its metrics.

Chapter 4 studies the single-encoder baselines and fixes the experimental frame, in a protocol-first order: an evaluation-protocol section first fixes the main metrics (QWK, macro-accuracy, macro-MAE), the development dataset (CMOSE), and the development loss (cross-entropy), each justified from up-front reasoning and in-domain evidence; the chapter then reads the in-domain leaderboard, studies prediction agreement to show the OpenFace and I3D families make different errors, and finally takes the baselines out of domain in the 3×3 matrix, closing the loop on the protocol.

Chapter 5 introduces the feature-group hybrid and tests it against that frame: it checks that the 486-configuration hybrid family repeats the protocol evidence of Chapter 4, answers the design questions (which encoder suits which group, and whether adding the I3D stream helps, an answer that depends on the test setting), and then takes the hybrid out of domain, where it outperforms the baseline in every cell of the 3×3 matrix and, with dataset combination, reaches the best result on the self-collected private set (RQ4).

Chapter 6 sums up the findings, restates the contributions in the light of the evidence, discusses the limits of the study, and proposes concrete directions

for future work, including targeting the rarest disengaged class, explicit domain adaptation, and group encoders that are learned rather than tried one by one.

CHAPTER 2. LITERATURE REVIEW

This chapter sets the technical and scientific context for the thesis. It first establishes the scope of the study (Section 2.1), then analyses the related work and its limitations (Section 2.2). The remaining sections present the foundational knowledge the thesis builds on and that later chapters assume: the feature extractors that turn a video into numerical descriptors (Section 2.3), the three families of temporal architecture used both as baselines and as the building blocks of the proposed model (Section 2.4), the loss functions for imbalanced ordinal targets (Section 2.5), the metrics used to evaluate ordinal imbalanced classification (Section 2.6), and the statistical measures used to compare and analyse the results (Section 2.7). The chapter closes by summarising how these pieces are combined in Chapter 3.

2.1 Scope of Research

This review is scoped to vision-based engagement recognition: classifying a learner’s engagement from facial video, where the only input is a frontal webcam recording of one person. Within this scope the thesis concentrates on four parts, and the review is organised around them: (i) extracting facial-behaviour and motion features, (ii) modelling the per-frame descriptors in time, (iii) loss functions suited to imbalanced ordinal labels, and (iv) the metrics used to evaluate such models.

Several adjacent problems are deliberately left out of scope. I do not estimate engagement from physiological sensors (heart rate, electrodermal activity, eye-tracking hardware), nor from interaction or clickstream logs, because they require instrumentation beyond a webcam. I do not treat general facial-expression or emotion recognition as an end in itself, referring to it only where it informs how engagement features are designed. Finally, the thesis works with OpenFace and I3D features at the descriptor level rather than re-training the extractors: CMOSE is the only dataset that ships these features precomputed, while for DAiSEE and the self-collected private set I extract them myself from the raw video using the same toolkits (Section 3.2); how those extractors are trained internally is reviewed for understanding but is not re-implemented.

2.2 Related Work

Engagement datasets and benchmarks. DAiSEE [2] is the most widely used in-the-wild benchmark for engagement recognition. It provides short webcam clips annotated with four ordinal engagement levels and is recorded under varied, uncontrolled conditions, which makes it realistic but noisy. CMOSE [1] is a more recent, larger, expert-labelled multimodal dataset that ships precomputed OpenFace

and I3D features (the only one of the three datasets used here to do so), and reports strong inter-rater reliability; its scale and label quality make it the thesis’s primary dataset. A systematic review of computer-vision engagement detection [8] surveys the common pipelines and names the two recurring open problems in the field, severe class imbalance and the subjectivity of engagement labels. This matches the motivation in Chapter 1.

Feature-based approaches. A large body of work first reduces each frame to a compact descriptor with a toolkit such as OpenFace [3] and then models the sequence. Representative pipelines feed OpenFace action units, gaze, and head pose to a recurrent network, for example an OpenFace+BiLSTM model [4], or apply dimensionality reduction and resampling (principal component analysis or singular value decomposition together with synthetic minority oversampling) before a convolutional or fully-connected classifier [9]. Other studies aggregate the features over time and compare classical learners such as logistic regression, support vector machines, multilayer perceptrons, k -nearest neighbours, and gradient-boosted trees [5]. These works show that OpenFace descriptors carry a real engagement signal, but they share a structural assumption: the 709-dimensional descriptor is treated as one uniform vector, encoded by one model with its temporal bias.

End-to-end video approaches. A second family learns directly from pixels. EfficientNetV2 [10] backbones followed by an LSTM [6] encode appearance per frame and integrate it over time, and context-aware affect-recognition models exploit spatial relationships in the scene [7]. These models can capture appearance and motion cues that landmark-based descriptors discard, but they are heavier to train, require the raw video rather than released features, and are less interpretable because the learned representation is not decomposed into nameable behavioural channels.

Research Gap. Two gaps persist across both families and define the contribution of this thesis. First, the facial features are mixed but are encoded as one block rather than per behavioural subsystem, even though their signals (gaze, eye and face geometry, head pose, action units) have very different timing. Second, studies mostly use accuracy to evaluate and do not have reliable evaluation for generalisation: in-domain accuracy dominates the literature, yet on imbalanced data a high accuracy score can hide a model that does not transfer at all. The feature-group hybrid proposed in Chapter 3 targets the first gap, and the 3×3 cross-dataset protocol of Chapter 5 targets the second.

2.3 Feature Extraction

2.3.1 OpenFace 2.0 facial-behaviour descriptors

OpenFace 2.0 [3] is an open-source facial-behaviour analysis toolkit that processes a video frame by frame and, for each frame, estimates four kinds of measurement: the two- and three-dimensional facial landmarks that outline the face and eye regions, the head pose (three translation and three rotation parameters) recovered from those landmarks, the eye-gaze direction vectors and angles for both eyes, and the intensity and presence of the facial action units of the Facial Action Coding System. Each frame is also tagged with a tracking-confidence and success flag, which downstream pipelines use to discard poorly tracked frames.

In CMOSE the per-frame descriptor used in this thesis is 709-dimensional. These 709 numbers are not homogeneous. By the meaning of their columns they split into five behavioural groups, gaze (8 dimensions), eye landmarks (280), face landmarks (340), head pose (46), and action units (35), and the thesis exploits this split throughout. These groups are physically distinct subsystems, each with its own dynamics: gaze and eye landmarks change rapidly and noisily; head pose drifts slowly and smoothly; action units fire in short, sparse bursts. Section 3.4 formalises this decomposition and Section 3.5 turns it into an architecture.

2.3.2 I3D motion and appearance features

The Inflated 3D ConvNet (I3D) [11] is a video representation obtained by “inflating” the two-dimensional convolutional filters of an ImageNet-pretrained image network into three dimensions, so that they operate over space and time jointly, and then pretraining the result on the large Kinetics dataset for action recognition. I3D therefore captures appearance and short-range motion that landmark-based descriptors omit. In the data used here, CMOSE ships a single precomputed 1024-dimensional I3D embedding per clip, and for DAiSEE and the private set I extract the same vector from the raw video. In every case it is one vector pooled over the whole clip in space and time, not a per-frame sequence. This thesis therefore treats I3D as a compact clip-level stream of motion and appearance that complements the OpenFace descriptors. The I3D stream contributes a global summary, not fine motion over time.

2.4 Temporal Architectures

A clip is a sequence of per-frame OpenFace descriptors, so a model must reduce a variable-length sequence to a single engagement prediction. Three architecture families dominate sequence modelling, and all three are used in this thesis, both

as single-encoder baselines (Section 3.3) and as the per-group encoders inside the hybrid (Section 3.5). They differ in how they let one time step influence another.

2.4.1 Temporal Convolutional Networks (TCN)

A Temporal Convolutional Network [12] applies one-dimensional convolutions along the time axis. Two design choices make it well suited to behavioural sequences. First, the convolutions are causal: the output at time t depends only on inputs at times $\leq t$, which is achieved by padding on the left and removing the surplus on the right. Second, successive layers use exponentially increasing dilation: a layer at depth l skips 2^l steps between the inputs it convolves, so a stack of L layers reaches a receptive field that grows as $\mathcal{O}(2^L)$ while keeping the number of parameters small. Each layer is wrapped in a residual block (two dilated convolutions with weight normalisation, a rectified-linear nonlinearity, and dropout, summed with a possibly projected copy of the input) that stabilises the training of deep stacks (Figure 2.1). The result is a strong inductive bias for short, local temporal motifs at multiple scales, which matches signals such as a head nod or an action-unit burst.

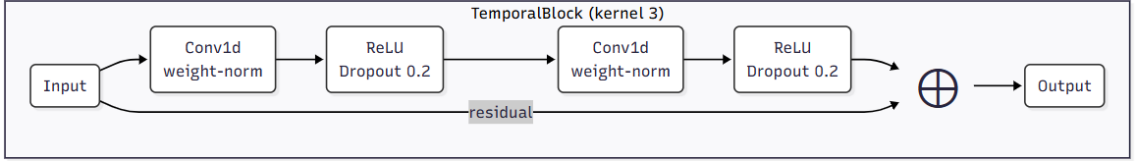


Figure 2.1: The residual TemporalBlock used in every TCN of this thesis: two weight-normalised dilated causal 1-D convolutions, each followed by a ReLU and dropout, summed with a residual connection (kernel size 3).

2.4.2 Long Short-Term Memory (LSTM)

The Long Short-Term Memory network [13] is a recurrent architecture that processes the sequence one step at a time while maintaining a cell state c_t and a hidden state h_t (Figure 2.2). Three gates (input i_t , forget f_t , and output o_t) regulate how much new information is written, how much past information is retained, and how much of the cell state is exposed:

$$\begin{aligned}
 f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\
 o_t &= \sigma(W_o x_t + U_o h_{t-1} + b_o), & \tilde{c}_t &= \tanh(W_c x_t + U_c h_{t-1} + b_c), \\
 c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, & h_t &= o_t \odot \tanh(c_t),
 \end{aligned} \tag{2.1}$$

where σ is the logistic sigmoid and $\odot$ is the element-wise product. The gating mechanism lets the network carry information across many time steps without the vanishing-gradient problem of plain recurrent networks, giving it a bias toward

longer-range dependencies. The cost is that the sequential recurrence is harder to parallelise than convolution or attention.

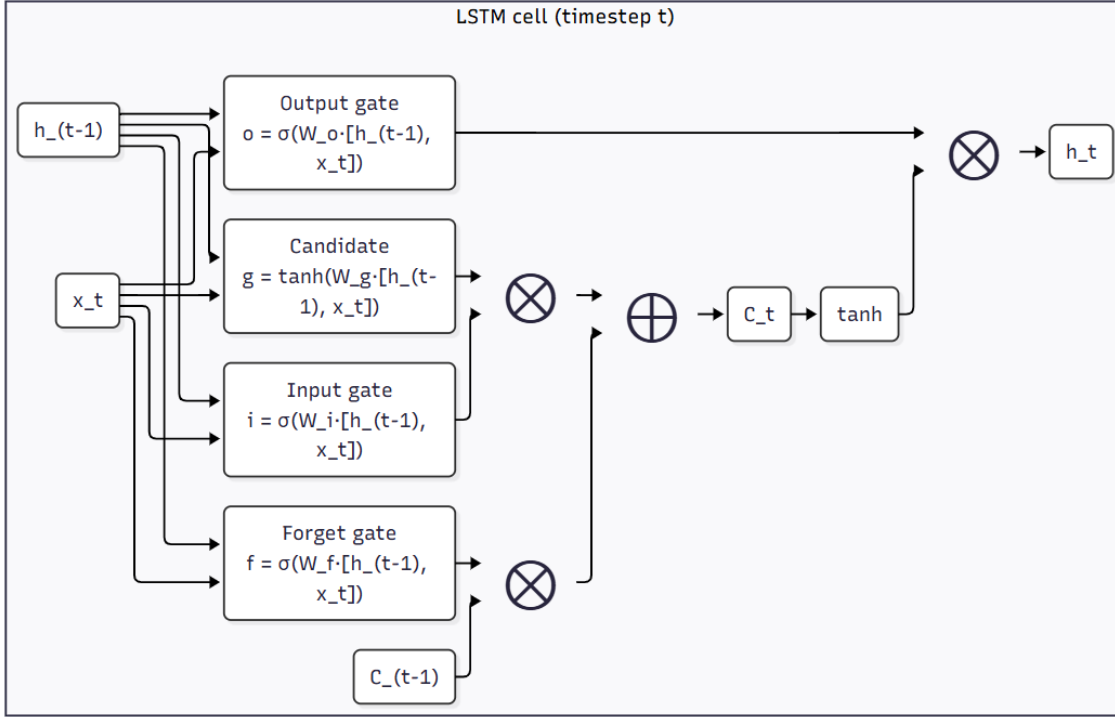


Figure 2.2: A single LSTM cell at time step t : the forget, input, and output gates regulate how the cell state C_t and hidden state h_t are updated from the previous states C_{t-1} , h_{t-1} and the input x_t .

2.4.3 Transformers

The Transformer encoder [14] drops recurrence and convolution and instead relates all time steps directly through self-attention. For queries Q , keys K , and values V obtained by linear projections of the input, attention computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (2.2)$$

so that every output position is a learned weighted combination of every input position, with d_k the key dimension. Because attention is permutation-invariant, order is injected through a positional encoding added to the input; this thesis uses the standard sinusoidal encoding. Multiple attention “heads” run in parallel and their outputs are concatenated, and each attention sub-layer is followed by a position-wise feed-forward network, with residual connections and layer normalisation throughout (Figure 2.3). The Transformer has the weakest temporal prior of the three families, since it can attend anywhere, which makes it flexible but reliant on more data than the other two.

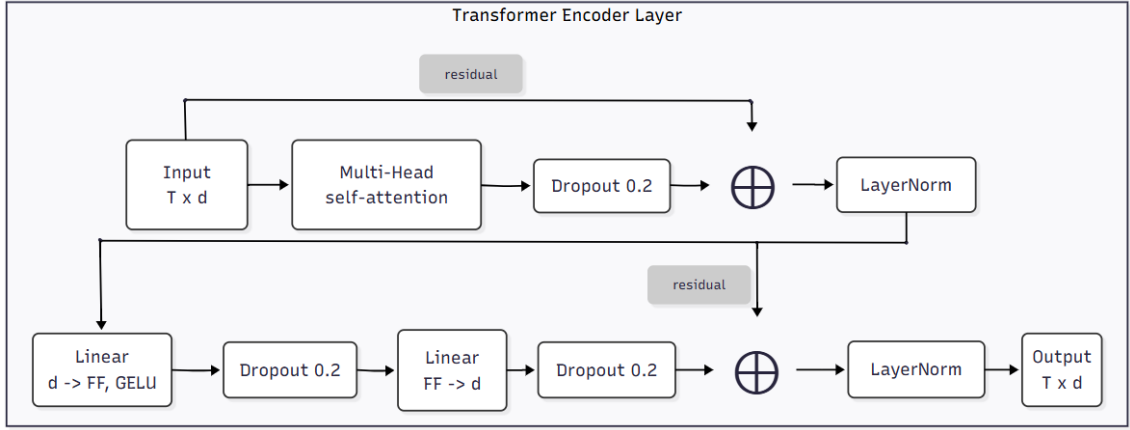


Figure 2.3: A single Transformer encoder layer used in this thesis: multi-head self-attention over the sinusoidally position-encoded input, followed by a position-wise feed-forward network, each sub-layer wrapped in a residual connection and layer normalisation.

2.5 Loss Functions for Imbalance and Ordinality

Let a model produce logits $z \in \mathbb{R}^C$ for the $C = 4$ engagement classes. The softmax function maps these logits to a probability distribution over the classes, the predicted probability of class c being

$$\hat{y}_c = \text{softmax}(z)_c = \frac{e^{z_c}}{\sum_{j=1}^C e^{z_j}} \quad (2.3)$$

so that $\hat{y}_c \in (0, 1)$ and $\sum_{c=1}^C \hat{y}_c = 1$. Let $y \in \{1, \dots, C\}$ be the ordinal ground-truth class, encoded as a one-hot vector through the indicator function $1[\cdot]$, which returns 1 when its argument (a logical statement) is true and 0 otherwise:

$$1[S] = \begin{cases} 1 & \text{if the statement } S \text{ is true} \\ 0 & \text{if } S \text{ is false} \end{cases} \quad (2.4)$$

so that the one-hot component of the label is $y_c = 1[c = y]$. Three loss families are compared in this thesis; their exact application (which weights, computed on which split, combined with which auxiliary term) is given in Section 3.6.

Cross-entropy (CE) is the standard classification loss,

$$\mathcal{L}_{\text{CE}} = - \sum_{c=1}^C y_c \log \hat{y}_c, \quad (2.5)$$

which, because y is one-hot, reduces to the negative log-probability of the true class; it treats all classes symmetrically and all errors as equally severe. On imbalanced

data it is dominated by the majority class and on ordinal data it ignores class order, so it serves as the baseline against which the other losses are judged.

Weighted cross-entropy counteracts imbalance by scaling each class’s contribution by a weight w_c inversely proportional to its frequency,

$$\mathcal{L}_{\text{WCE}} = - \sum_{c=1}^C w_c y_c \log \hat{y}_c. \quad (2.6)$$

This raises the cost of mistakes on rare classes such as Highly-Disengage and forces the model to avoid collapsing onto the majority, at the expense of some overall accuracy. Resampling techniques such as SMOTE [15] address the same imbalance at the data level by synthesising minority examples; this thesis instead studies loss-level remedies, which require no change to the data pipeline.

Ordinal (squared-EMD) loss [16] encodes the order of the classes by comparing cumulative distributions. Writing the predicted cumulative distribution as $\sum_{j=1}^k \hat{y}_j$ and the cumulative distribution of the one-hot label as the unit step $\sum_{j=1}^k y_j$ (which rises from 0 to 1 at the true class), the loss is the mean squared distance between the two step functions,

$$\mathcal{L}_{\text{EMD}} = \frac{w_y}{C} \sum_{k=1}^C \left(\sum_{j=1}^k \hat{y}_j - \sum_{j=1}^k y_j \right)^2, \quad (2.7)$$

the squared Earth Mover’s Distance (squared-EMD, also called the CDF-MSE) between the predicted and target distributions over the ordered classes [16], not the exact Earth Mover’s (1-Wasserstein) distance, which is the unsquared L_1 distance between the same cumulative distributions; the leading w_y is the weight of the true class (Section 3.6), set to 1 in the unweighted variant. Because moving probability mass across several classes costs more than moving it to an adjacent class, this loss directly penalises the large ordinal errors (HD $\leftrightarrow$ HE) that accuracy and plain cross-entropy treat as ordinary.

All models are optimised with Adam [17], an adaptive first-order optimiser that maintains per-parameter running estimates of the first and second moments of the gradient and is the standard default for training such networks.

2.6 Evaluation Metrics

Because the labels are ordinal and imbalanced, the choice of metric is itself a research question (RQ4). This thesis reports six metrics over a test set of N samples with true labels y_i and predicted labels $\hat{y}_i$ (for $i = 1, \dots, N$), but treats exactly one

of them, QWK, as the primary metric that selects the best model; the other five are secondary, reported to describe a result but never to choose one. Where a confusion matrix is needed, O_{ij} denotes the number of samples of true class i predicted as class j .

Accuracy is the fraction of correct predictions,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i = \hat{y}_i]. \quad (2.8)$$

It is reported for comparability with the literature but is explicitly not a primary metric here, because on imbalanced data it rewards majority prediction.

Macro-accuracy (balanced accuracy) is the mean of the per-class recalls,

$$\text{MacroAcc} = \frac{1}{C} \sum_{c=1}^C \frac{\sum_{i=1}^N \mathbf{1}[y_i = \hat{y}_i = c]}{\sum_{i=1}^N \mathbf{1}[y_i = c]}, \quad (2.9)$$

giving every class equal weight regardless of frequency. It exposes the minority-class collapse that accuracy hides; here it is a secondary metric, reported to characterise class balance but never used to select a model.

Mean absolute error (MAE) treats the class indices as an ordinal scale and averages the distance between truth and prediction,

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|, \quad (2.10)$$

while macro-MAE averages the per-class MAE so that errors on rare classes are not drowned out,

$$\text{MacroMAE} = \frac{1}{C} \sum_{c=1}^C \frac{\sum_{i=1}^N \mathbf{1}[y_i = c] |y_i - \hat{y}_i|}{\sum_{i=1}^N \mathbf{1}[y_i = c]}. \quad (2.11)$$

Both reward predictions that are at least close on the scale even when not exactly right; both are smaller-is-better. Macro-MAE is a secondary metric: it is the natural ordinal-distance companion to QWK, reporting how far a prediction misses rather than only whether it misses, with each class weighted equally so the rare disengaged classes are not masked by the majority. Like the other secondary metrics, though, it is reported alongside QWK and never used to select a model.

Cohen's kappa [18] measures agreement corrected for the agreement expected by chance. With observed agreement p_o and expected agreement p_e (computed from

the marginals), $\kappa = (p_o - p_e)/(1 - p_e)$. In its weighted form it is written as

$$\kappa = 1 - \frac{\sum_{ij} w_{ij} O_{ij}}{\sum_{ij} w_{ij} E_{ij}}, \quad (2.12)$$

where E_{ij} is the expected count under independent marginals and w_{ij} is a penalty matrix. The unweighted variant uses $w_{ij} = 1$ off the diagonal and 0 on it.

Quadratic-weighted kappa (QWK) is the same agreement coefficient with the quadratic penalty

$$w_{ij} = \frac{(i - j)^2}{(C - 1)^2}, \quad (2.13)$$

so that disagreements are penalised in proportion to the square of the ordinal distance between the true and predicted classes. QWK suits this problem well because it is corrected for chance, it respects the class order, and it cannot be inflated by majority prediction. It is the thesis's primary metric, the only criterion used to select the best model, with macro-accuracy (balanced recall) and macro-MAE (ordinal distance) reported as secondary companions that describe a result but never choose it.

2.7 Statistical Measures for Result Analysis

Beyond scoring each model in isolation, the results chapters ask three comparative questions that need their own statistical tools: do the six metrics rank the configurations the same way (Section 4.2.1), does in-domain strength predict cross-dataset strength (Section 4.5.3), and do different models make the same per-clip decisions (Section 4.4)? The first two are rank-correlation questions; the third is an agreement question.

Spearman's rank correlation ρ [19] measures how well a monotonic relationship describes two variables by applying the Pearson correlation coefficient to their ranks rather than their values. For n paired observations with rank differences d_i and no ties it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}, \quad (2.14)$$

ranging from -1 (perfectly inverted ordering) through 0 (no monotonic relation) to $+1$ (identical ordering). This thesis uses ρ to test whether a model's in-domain QWK predicts its QWK on unseen target datasets.

Kendall's τ [20] measures rank agreement through concordance rather than squared rank distance. Counting the pairs of observations ordered the same way in both variables (concordant, n_c) against those ordered oppositely (discordant, n_d),

and assuming no ties, the coefficient is

$$\tau_a = \frac{n_c - n_d}{\binom{n}{2}}, \quad (2.15)$$

again bounded by $[-1, 1]$. It is less sensitive to small samples and to a few large rank displacements than ρ , which is why it is used here to quantify how far the six metrics agree on the ranking of the baseline configurations.

Prediction agreement and ensemble headroom. To ask whether two models agree clip by clip rather than merely score alike, the thesis reuses Cohen’s κ (Section 2.6) between the two models’ predicted-label vectors; chance-correction matters because two models that lean toward the majority class agree often by default. The headroom available from combining a pool of M configurations is bounded by the oracle accuracy, the fraction of clips that at least one configuration classifies correctly,

$$\text{OracleAcc} = \frac{1}{N} \sum_{n=1}^N \mathbf{1} \left[\exists m \in \{1, \dots, M\} : \hat{y}_n^{(m)} = y_n \right], \quad (2.16)$$

which upper-bounds any fixed combiner, and is contrasted with the realistic majority-vote accuracy obtained by taking the modal prediction across the pool. A large gap between the best single model, the majority vote, and the oracle indicates that the models err on different clips and that an ensemble has room to improve.

2.8 Chapter Summary

This chapter scoped the thesis to ordinal engagement recognition from a webcam and analysed the two main approaches in the literature, identifying their shared limits: mixed facial descriptors are encoded as one block, and transfer across datasets is rarely measured with metrics that resist majority-class inflation. It then laid the technical foundations the rest of the thesis depends on: the OpenFace 709-dimensional descriptor and its five behavioural groups, the clip-level I3D embedding, the three families of temporal architecture (TCN, LSTM, Transformer) and their different temporal biases, the three loss families for imbalance and ordinality, the six metrics with QWK as the primary metric and macro-accuracy and macro-MAE as secondary metrics, and the rank-correlation (Spearman ρ , Kendall τ) and prediction-agreement statistics used to analyse the results. Chapter 3 combines these foundations into a concrete methodology: it turns the five-group decomposition into the proposed feature-group hybrid, specifies the baselines and the exact loss forms, and defines the training and 3×3 cross-dataset protocols.

CHAPTER 3. METHODOLOGY

This chapter specifies the methodology of the thesis. Section 3.1 gives an overview of the complete pipeline as a single diagram. Section 3.2 describes the three training datasets, the private test set I collected, and the preprocessing that makes them comparable. Section 3.3 defines the single-encoder baselines that anchor every comparison. Section 3.4 then formalises the feature-group decomposition of the OpenFace descriptor, and Section 3.5 builds the proposed feature-group hybrid on top of it, in explicit contrast to those baselines. Section 3.6 establishes the exact loss forms, Section 3.7 the training protocol, and Section 3.8 the 3×3 cross-dataset protocol and its metrics.

3.1 Overview

The task is framed as ordinal classification of a fixed-length clip into four classes. Per-frame OpenFace descriptors and a clip-level I3D embedding are extracted, brought to a common temporal length, and normalised. The single-encoder baselines (Section 3.3) follow the standard approach: a single encoder consumes the full 709-dimensional OpenFace sequence (or the pooled I3D vector) and feeds a classifier. The proposed model replaces that single encoder with a split-and-encode design: the OpenFace descriptor is split into five feature groups, each encoded independently into a 64-dimensional embedding by its own temporal encoder; in the multimodal variant the I3D clip vector is encoded into a 128-dimensional embedding by a small MLP. The embeddings are concatenated and classified by a shared multilayer perceptron, while an auxiliary head supervises every stream separately; the hybrid is shown in Figure 3.3. Every model is trained and evaluated under a 3×3 cross-dataset protocol and three loss functions.

3.2 Data and Preprocessing

3.2.1 Datasets

The thesis uses three training sources and three test sets, whose sizes and class balance are reported in Table 3.1.

- (i) **CMOSE** [1]: 12,197 clips, divided into a `train` split (8,783), an `unlabel` split (2,193) that this thesis uses as the validation set for early stopping and checkpoint selection, and a `test` split (1,221).
- (ii) **DAiSEE** [2]: 8,571 clips, whose official `Train/Validation/Test` partitions are mapped onto the same `train/unlabel/test` roles (5,358 / 1,429 / 1,784).

- (iii) **Combined**: the combination of CMOSE and DAiSEE (20,768 clips), with source-prefixed identifiers so the two datasets cannot collide. All three splits are merged split-wise, giving 14,141 / 3,622 / 3,005 train / validation / test clips.
- (iv) **Private (self-collected)**: a held-out set of 366 clips that I collected and manually labelled myself, used for testing only, never in training or model selection. Its construction is described in Section 3.2.2.

All four sets share the four ordinal labels HD/DE/EG/HE. Table 3.1 and Figure 3.1 show that all three training datasets are heavily imbalanced toward the upper-middle classes, and imbalanced in different directions. CMOSE is dominated by Engage (69.4%) with only 2.8% Highly-Disengage, whereas DAiSEE is split between Engage (50.3%) and Highly-Engage (43.8%) with just 0.7% Highly-Disengage. The rare disengaged classes are precisely the ones a monitoring system must catch, yet they amount to a fraction of a percent to a few percent of the data. The same imbalance holds within each of the train, validation, and test partitions, so it is the nature of the problem, not created by how the data was split.

Table 3.1: Dataset statistics and class balance (%).

Dataset	Total	Train	Val	Test	HD%	DE%	EG%	HE%
CMOSE	12197	8783	2193	1221	2.800	18.100	69.400	9.600
DAiSEE	8571	5358	1429	1784	0.700	5.100	50.300	43.800
Combined	20768	14141	3622	3005	2.000	12.800	61.500	23.700

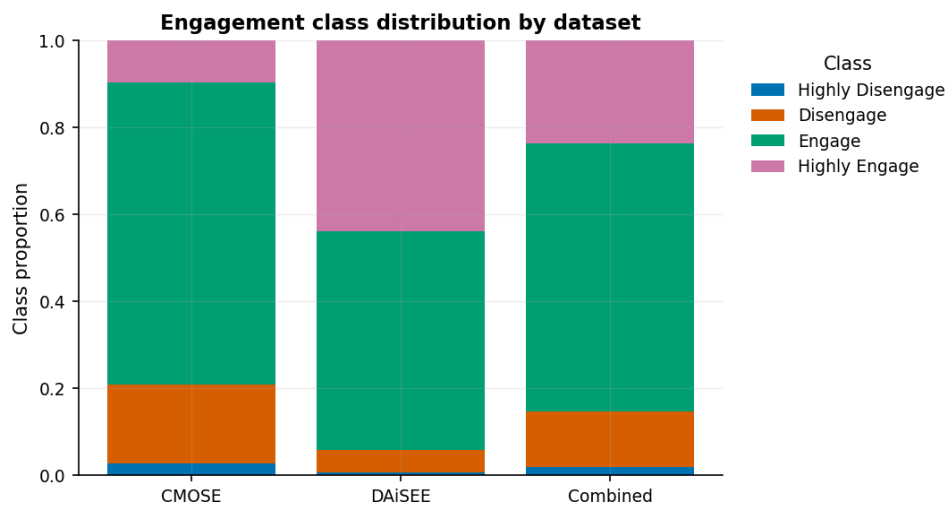


Figure 3.1: Engagement class distribution by dataset.

This severe and uneven imbalance shapes three choices the thesis returns to. It predicts difficulty in transferring between datasets, where a model fit to one label

prior meets a sharply different one at test time (Chapters 4 and 5); it motivates the imbalance-aware losses of Section 3.6; and, because predicting the majority class alone scores well on every dataset, it rules out raw accuracy as a selection criterion in favour of quadratic-weighted kappa (QWK), the single metric on which every best model in this thesis is chosen, with macro-accuracy and macro-MAE kept only as secondary balanced checks (Section 3.8).

3.2.2 The private evaluation set

The private set measures deployment rather than in-dataset behaviour; because no model is ever trained or selected on it, it is a test-only set. Although it is small, it is sampled in the wild, from the kind of webcam recordings the model will face in real deployment. It also serves a purpose specific to engagement. The OpenFace gaze, head-pose, and landmark features are measured in the camera’s image geometry, so a model validated only on the narrow range of recording setups in a public dataset offers no guarantee on the varied webcam angles of a real deployment. A small, locally collected, hand-labelled test set is the most direct way to check that a model is fit for its intended setting before deployment. It was built in five steps.

- (i) **Source and segmentation.** Webcam recordings of online-class sessions were sampled at several points within each session and cut into fixed-length 10-second clips, each containing a single learner.
- (ii) **Tracking-quality filtering.** Every clip was run through the same OpenFace pipeline as the public datasets; clips with too few tracked frames were rejected automatically, since a descriptor built from unreliable frames is meaningless.
- (iii) **Manual labelling.** The accepted clips were labelled by me through a purpose-built local interface, one engagement label per clip; reliance on a single annotator is a limitation revisited in Section 6.4.
- (iv) **Feature extraction.** Accepted clips were processed through the same OpenFace (709-d per frame) and I3D (1024-d per clip) extraction pipeline I applied to the raw DAiSEE video, identical in format to the features CMOSE ships precomputed, so the private features are distributionally comparable to the training features.
- (v) **Final set.** Of the labelled clips, the 366 with complete, aligned OpenFace and I3D features form the private evaluation set. Its label prior differs again from both public datasets (58.2% Engage, 30.6% Highly-Engage, 8.5% Disengage, and only 2.7% Highly-Disengage), which, together with its independent

recording conditions, makes it an in-the-wild measure of how the models would behave in deployment. It anchors Section 5.6.

3.2.3 Features and preprocessing

Features. Each clip is represented by per-frame OpenFace 2.0 descriptors (709-d) [3] and a clip-level I3D embedding (1024-d) [11]. CMOSE is the only dataset that ships these features precomputed; for DAiSEE and the private set I extract both from the raw video with the same toolkits, so all three sources share an identical feature format. When a single OpenFace frame contains more than one detected face, the highest-confidence row is kept; non-numeric entries are coerced to zero.

Temporal resampling. OpenFace clips vary in length, so each is resampled along time to a fixed number of frames by linear interpolation per feature. The OpenFace sequence length is 300 frames for the single-dataset runs and is reduced to 150 for the larger Combined runs to fit memory; in the multimodal variant the OpenFace stream is resampled to 75 frames. The I3D feature is a single 1024-dimensional vector per clip, not a sequence, and is encoded by an MLP (Section 3.5).

Normalisation. Every feature is standardised by a per-feature z -score whose mean and standard deviation are fit on the training split only; the validation, test, and private sets are transformed with those training statistics, so no target-set information leaks into preprocessing. OpenFace and I3D are standardised with separate statistics.

3.3 Baseline Models

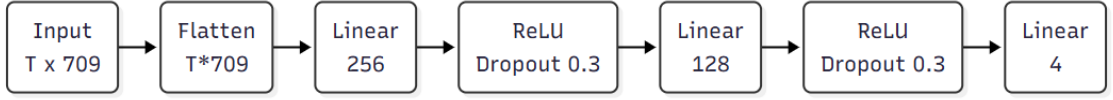
Five single-encoder baselines provide the comparison points; they form the allow-list `MODEL_ORDER` used by the analysis. Each follows the standard approach: a single encoder consumes the full feature sequence and feeds a classifier.

- (i) `openface_mlp`: flattens the OpenFace sequence and classifies it with a multilayer perceptron ($\rightarrow 256 \rightarrow 128 \rightarrow 4$, dropout 0.3).
- (ii) `openface_tcn`: a TCN over the 709-d sequence (residual dilated blocks of widths 256, 128, 128; kernel 3; dilations 1, 2, 4), adaptive average-pooling, then a $128 \rightarrow 128 \rightarrow 4$ classifier.
- (iii) `openface_lstm`: a two-layer LSTM (hidden width 256) whose final hidden state feeds a $256 \rightarrow 128 \rightarrow 4$ classifier.
- (iv) `openface_transformer`: a linear projection to 128 dimensions, a sinusoidal positional encoding, and two Transformer encoder layers (4 heads, feed-forward width 256), mean-pooled and classified.

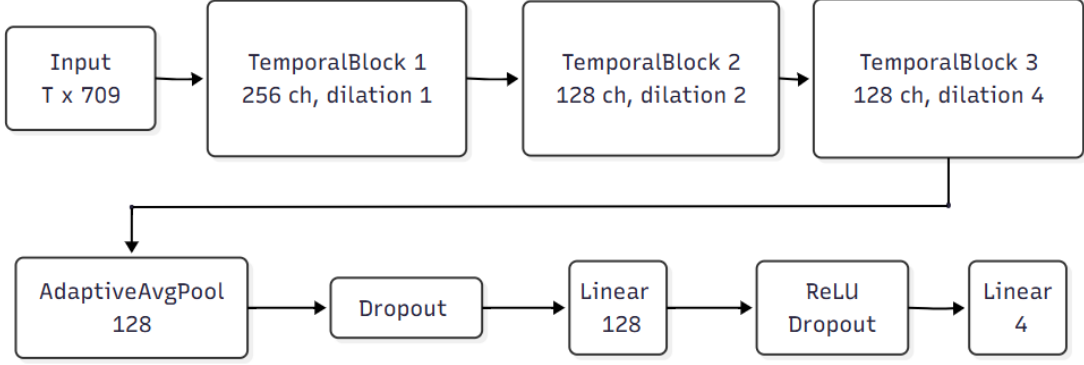
- (v) `i3d_mlp`: classifies the single 1024-d I3D clip vector with an MLP ($1024 \rightarrow 256 \rightarrow 128 \rightarrow 4$, dropout 0.3).

A sixth model, `openface_tcn_i3d_fusion` (a shared-fusion TCN over both modalities), exists in the codebase but is excluded from the assessment so that the comparison isolates the feature-group hypothesis rather than modality fusion.

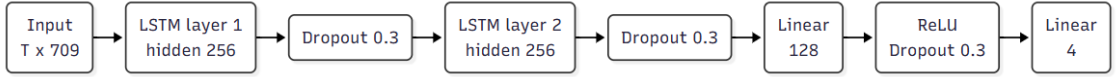
The five baseline architectures are shown in Figure 3.2. The four OpenFace baselines (a–d) encode the full $T \times 709$ sequence with a single encoder, while `i3d_mlp` (e) classifies the pooled 1024-d I3D clip vector. Forcing one encoder, and therefore one temporal bias, onto all 709 mixed features at once is the design decision the proposed model revisits in Sections 3.4 and 3.5.



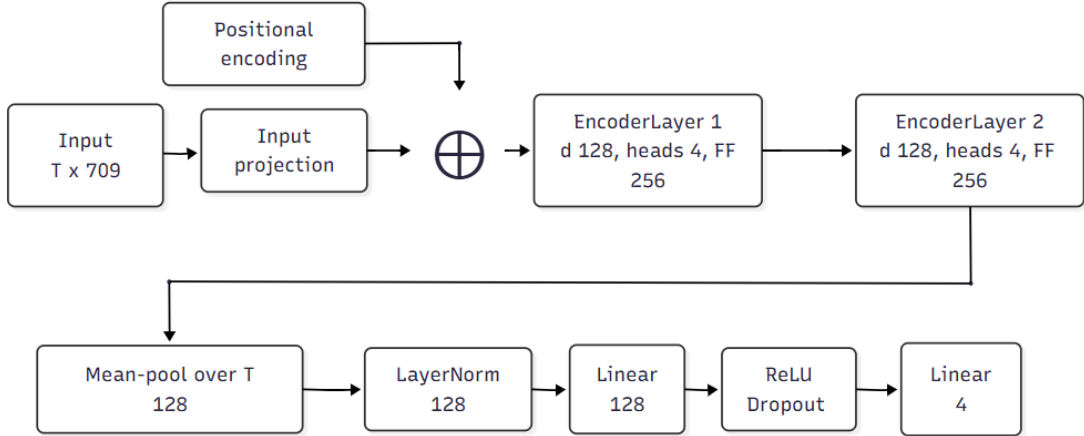
(a) `openface_mlp`: the $T \times 709$ sequence is flattened and classified by an MLP.



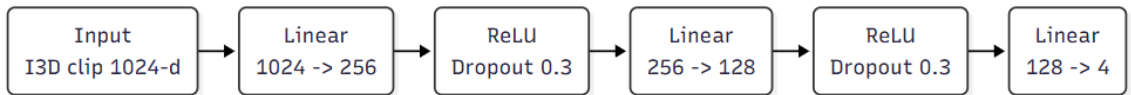
(b) `openface_tcn`: a TCN over the 709-d sequence, adaptively average-pooled and classified.



(c) `openface_lstm`: a two-layer LSTM over the 709-d sequence whose final state is classified.



(d) `openface_transformer`: a Transformer encoder over the 709-d sequence, mean-pooled and classified.



(e) `i3d_mlp`: the pooled 1024-d I3D clip vector is classified by an MLP.

Figure 3.2: The five single-encoder baselines. Each applies a single encoder to the full feature sequence (or pooled I3D vector) before classification.

3.4 Grouping the Facial Features by Meaning

The 709 OpenFace features are partitioned into five meaningful groups by matching the column names, in the fixed order used everywhere in the pipeline (gaze, eye, face, head, action units). Table 3.2 summarises the partition; every one of the 709 columns belongs to exactly one group.

Table 3.2: Grouping of the 709-dimensional OpenFace descriptor into five feature groups by meaning.

Group	Content	Dim
Gaze	gaze direction vectors and angles	8
Eye landmarks	2-D/3-D eye-region landmark coordinates	280
Face landmarks	2-D/3-D facial landmark coordinates	340
Head pose	head translation and rotation parameters	46
Action units	facial action-unit presence and intensity	35
Total		709

This decomposition is the inductive prior of the method. The five subsystems have different dynamics, so each can in principle be matched to an encoder whose temporal bias suits it. Gaze and eye landmarks show rapid, noisy micro-saccades; head pose drifts slowly and smoothly; action units fire in brief, sparse bursts. Encoding all 709 features together, as the single-encoder baselines of Section 3.3 do, forces a single bias on all of them at once.

3.5 Hybrid Architecture (Proposed)

The proposed model, shown in Figure 3.3, encodes each stream independently, fuses the resulting embeddings, and supervises every stream with its own auxiliary head. It is built from the same three families of encoder as the OpenFace baselines of Section 3.3 (TCN, Transformer, and LSTM), but where a baseline applies one encoder to the full $T \times 709$ sequence, the hybrid assigns an independently chosen encoder to each of the five feature groups of Section 3.4, so any difference in performance can be attributed to the decomposition rather than to a different kind of encoder.

Per-group encoder. Each group is encoded by a `GroupEncoder` that is one of three interchangeable temporal encoders (Figure 3.4), followed by temporal mean-pooling and a layer normalisation, producing a 64-dimensional embedding:

- (i) a TCN: two residual blocks of dilated causal 1-D convolutions (channel widths 64, 64; kernel size 3; dropout 0.2);

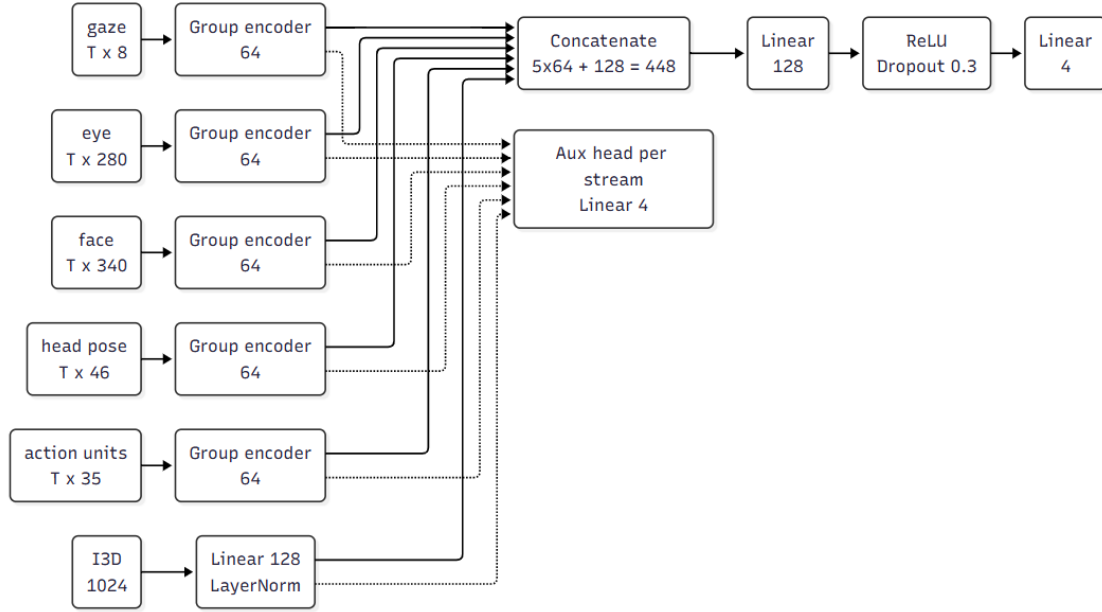
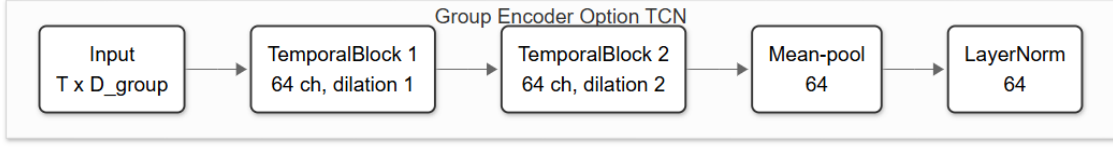


Figure 3.3: Feature-group hybrid architecture. Each OpenFace group is encoded independently into a 64-d embedding (group encoders are chosen per group from TCN / Transformer (T) / LSTM); the I3D clip vector is encoded by an MLP into a 128-d embedding. The six embeddings are concatenated (448-d) and classified by a shared MLP; each stream also feeds an auxiliary head. The OpenFace-only variant drops the I3D row and fuses a 320-d vector.

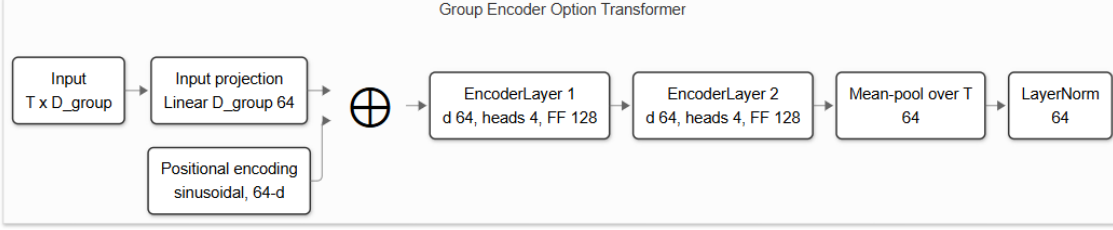
- (ii) a Transformer encoder: a linear projection to 64 dimensions, a sinusoidal positional encoding, and two encoder layers (4 heads, feed-forward width 128, GELU, dropout 0.2);
- (iii) an LSTM: a two-layer recurrent encoder of hidden width 64 (dropout 0.2).

Fusion and heads. The five 64-d group embeddings are concatenated into a 320-d vector and passed through a two-layer MLP classifier ($320 \rightarrow 128 \rightarrow 4$, ReLU, dropout 0.3) that produces the main logits. Each group additionally feeds an auxiliary classification head (dropout $\rightarrow$ Linear($64 \rightarrow 4$)). The total loss is the main-head loss plus $\text{aux_weight} = 0.2$ times the mean of the per-group losses (Section 3.6); the auxiliary supervision makes each stream individually discriminative.

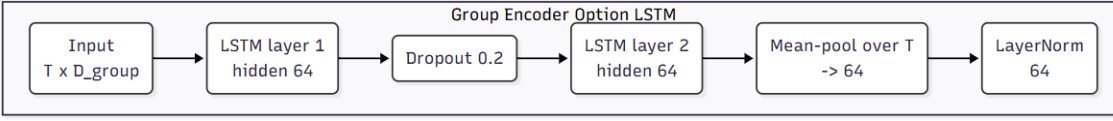
Multimodal variant. The I3D-fused model adds a sixth stream: the single 1024-d I3D clip vector is encoded by an MLP ($1024 \rightarrow 256 \rightarrow 128$, ReLU, dropout 0.3) and layer-normalised to a 128-d embedding with its own auxiliary head. The fusion MLP then takes a 448-d input ($5 \times 64 + 128$). The I3D stream uses an MLP rather than a temporal encoder because the provided I3D feature is a single pooled vector, not a sequence.



(a) TCN group encoder.



(b) Transformer group encoder.



(c) LSTM group encoder.

Figure 3.4: The three interchangeable per-group encoders. Each maps a group’s $T \times D_{\text{group}}$ sequence to a 64-dimensional embedding through temporal mean-pooling and layer normalisation; one is chosen independently for each of the five groups.

Configuration and arch_key. A hybrid configuration is fully specified by the encoder chosen for each of the five groups, written as an `arch_key` of five tokens in group order, for example `TCN_T_TCN_LSTM_T` ($T = \text{Transformer}$, $\text{TCN} = \text{TCN}$, $\text{LSTM} = \text{LSTM}$). With three choices per group there are $3^5 = 243$ configurations, each evaluated with and without the I3D stream (Section 5.2).

3.6 Loss Functions

The five baseline models are each trained under all three losses introduced in Section 2.5 (cross-entropy, weighted cross-entropy, and the ordinal squared-EMD loss of Equation (2.7)), so that the effect of the loss can be measured directly. The class weights for the weighted and ordinal variants are inverse-frequency weights computed on the training split and normalised to mean 1: for class c with n_c training samples and C non-empty classes,

$$w_c = \frac{1}{n_c} \cdot \frac{C}{\sum_k \frac{1}{n_k}}, \quad \text{so that} \quad \frac{1}{C} \sum_c w_c = 1. \quad (3.1)$$

The ordinal loss is applied in its class-weighted form as well, multiplying each sample’s CDF distance by w_y .

For the hybrid models the objective also includes the auxiliary term. With main logits z and per-stream auxiliary logits $\{a^{(s)}\}$, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{main}}(z, y) + \lambda_{\text{aux}} \cdot \frac{1}{S} \sum_{s=1}^S \mathcal{L}_{\text{CE}}(a^{(s)}, y), \quad (3.2)$$

where $\mathcal{L}_{\text{main}}$ is the main-head classification loss, $\lambda_{\text{aux}} = 0.2$ is the auxiliary weight, and the auxiliary heads are always trained with plain cross-entropy, over the S streams (five groups, or six with I3D). The formulation admits any of the three losses for the main head; the 243-configuration hybrid ablation reported here fixes $\mathcal{L}_{\text{main}}$ to cross-entropy, so that its results isolate the effect of the architecture from that of the loss.

3.7 Training Protocol

All models are optimised with Adam [17] at a learning rate of 1×10^{-4} . Training uses mini-batches (128 for the baseline runs, 64 for the 243-configuration ablation) and early stopping on the validation loss (the CMOSE `unlabel` split, and its analogues for DAiSEE and Combined) with a patience of 10 epochs and an epoch cap (800 for baselines, 400 for the ablation). The checkpoint with the lowest validation loss is restored for testing. Following the convention adopted in this project, only the training and validation loss are recorded per epoch; the final metrics are evaluated exactly once, on the held-out test split. Runs use a fixed random seed (42) and optional automatic mixed precision on GPU, and every run converges stably and early-stops well before the cap, so the differences reported between models are architectural rather than optimisation artefacts.

3.8 Evaluation Protocol

Cross-dataset matrix. Generalisation is measured with a 3×3 matrix: every model and loss is trained on each of {CMOSE, DAiSEE, Combined} and evaluated on each of {CMOSE-test, DAiSEE-test, Private}. The diagonal cells are the in-domain results; the off-diagonal and the Private column measure transfer to unseen distributions.

Metrics. Each evaluation reports the six metrics defined in Section 2.6 (accuracy, macro-accuracy, MAE, macro-MAE, Cohen’s kappa, and QWK), but QWK is the single primary metric: it alone selects the best model. The other five are secondary, reported for analysis and for comparability with the literature, but never used to pick a winner. The empirical justification for selecting on QWK alone is laid out in the evaluation-protocol section of Chapter 4 (Section 4.2). Row-normalised confusion matrices and per-class F1 scores are used for error analysis.

Aggregation and reproducibility. The complete study comprises roughly 1,500 trained runs (the five baselines $\times$ three losses $\times$ three sources, and the 243 per-group hybrid configurations with and without I3D); all are aggregated programmatically, and every figure and table of Chapters 4 and 5 is produced from the aggregate by a single command.

3.9 Chapter Summary

This chapter specified the methodology end to end: the pipeline; the three training datasets and their imbalance, the test-only private set and its collection, and the leakage-free preprocessing; the five single-encoder baselines; the five-group decomposition of the OpenFace descriptor; and the feature-group hybrid (Figure 3.3) that reuses the baselines' encoder families per group, with its fusion and auxiliary heads, I3D variant, and 243-configuration design space. It also established the loss forms, the training protocol, and the 3×3 evaluation protocol with its single primary metric (QWK) and its secondary metrics, which Chapters 4 and 5 apply to the in-domain and generalisation results.

CHAPTER 4. BASELINE MODELS

4.1 Overview

This chapter studies the single-encoder baselines (a feature sequence pushed through a single temporal encoder, Section 3.3) and establishes the methodological frame the rest of the thesis depends on. Section 4.2 establishes the evaluation protocol: the metrics, the development dataset, and the development loss; the two arguments that require out-of-domain evidence (that accuracy cannot detect cross-dataset collapse, and that the loss ranking is stable under shift) are stated there as claims and proved in Section 4.5.2. Under the established protocol, Section 4.3 ranks the five baselines and sets the reference level the proposed architecture must exceed; Section 4.4 moves from scores to individual predictions and asks whether models with similar scores give the same answers, the analysis that motivates the hybrid design of Chapter 5; and Section 4.5 takes the baselines out of domain with the 3×3 cross-dataset matrix.

How results are aggregated. Two conventions hold in both results chapters. A winner is the single configuration that attains the highest QWK over the tunable axis (the loss, and in Chapter 5 the per-group encoder assignment): QWK is the only selection metric, so a reported “best” model never depends on any other number, and ties are broken stable-first with no secondary metric consulted. A claim about a family of configurations is instead reported as means and distributions. Every configuration is trained once with a fixed seed (Section 3.7), so a reported mean is a mean over the design grid: robustness to design choices, not a noise estimate across seeds.

4.2 Evaluation Protocol

4.2.1 Metrics

Engagement is an ordinal, severely imbalanced four-class problem: the levels lie on a line, and 69.4% of CMOSE is the single class Engage (Table 3.1). Raw accuracy ignores both properties (it rewards majority-class collapse and counts a one-level miss the same as a three-level miss), whereas QWK is chance-corrected and weights every error by the squared distance between predicted and true levels (Section 2.6). On first principles, accuracy is the wrong metric for this task, and the in-domain data confirm this in two ways.

First, the leaderboard winner depends on the metric. Table 4.1 reports which of the fifteen baseline configurations (five architectures $\times$ three losses) each of the

six metrics declares best in-domain on CMOSE, and no two camps agree: QWK picks the OpenFace TCN under cross-entropy (0.537), both macro metrics pick the I3D MLP under the ordinal loss (macro-accuracy 0.635, macro-MAE 0.458), and accuracy, micro-MAE, and Cohen’s kappa all pick the I3D MLP under cross-entropy (0.773, 0.244, 0.436). The choice of metric changes the result, so it must be argued rather than left to default.

Table 4.1: The winning base configuration according to each metric (in-domain CMOSE).

Metric	Best model	Loss	Value
QWK	openface_tcn	CE	0.537
Macro-Accuracy	i3d_mlp	Ordinal	0.635
Macro-MAE	i3d_mlp	Ordinal	0.458
Accuracy	i3d_mlp	CE	0.773
MAE	i3d_mlp	CE	0.244
Cohen κ	i3d_mlp	CE	0.436

Second, the six metrics form two blocks, bridged only by QWK. Figure 4.1 reports the Kendall rank correlation between every pair of metrics over the fifteen configurations: a micro block (accuracy and micro-MAE, $\tau = 0.98$, dominated by the majority class) and a macro block (macro-accuracy and macro-MAE, $\tau = 0.89$) that are almost unrelated to each other (cross-block $\tau \approx 0.01$ – 0.10). QWK is the only metric with moderate-to-strong agreement with both blocks ($\tau = 0.49$ – 0.60): the compromise metric, sensitive to ordinal distance without being dominated by the majority class.

Protocol. The thesis therefore adopts QWK as its single primary metric, the only criterion that selects the best model: the winner is the argmax of QWK, and a tie on QWK is broken stable-first, with no second metric ever consulted. The remaining five metrics (macro-accuracy, macro-MAE, accuracy, micro-MAE, and Cohen’s kappa) are secondary: they are reported alongside every result to describe it (the macro pair tracks class balance and ordinal error, accuracy and micro-MAE allow comparison with the literature), but none of them ever selects a model. Deferred evidence: the decisive argument against accuracy (that it cannot even detect the collapse of a model’s ordinal signal under distribution shift) is delivered in Section 4.5.2.

4.2.2 Development Corpus

Table 4.2 compares the two in-domain cells of the evaluation matrix: the best in-domain model reaches QWK 0.537 on CMOSE but only 0.166 on DAiSEE

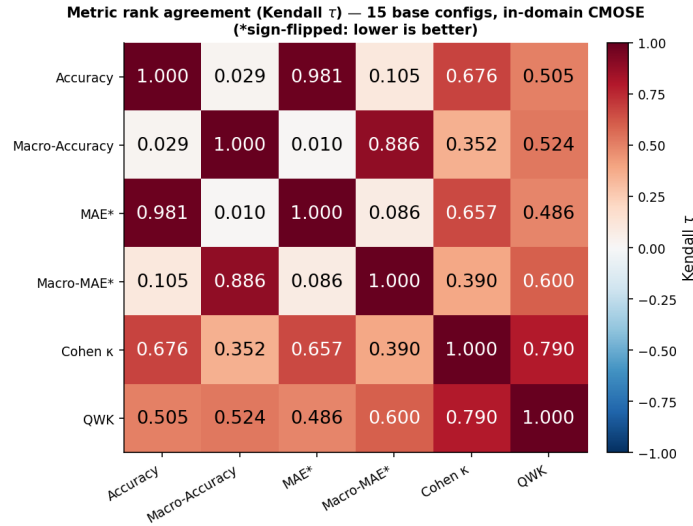


Figure 4.1: Rank agreement (Kendall τ) between the six metrics over the fifteen baseline configurations, in-domain on CMOSE (MAE metrics sign-flipped so positive τ means agreement). The metrics form a micro block and a macro block that barely agree; QWK is the only metric correlated with both.

(macro-accuracy 0.535 versus 0.289, macro-MAE 0.547 versus 1.226, more than twice the ordinal error), while DAiSEE’s raw accuracy of 0.548 hides the failure, a first sign of how misleading accuracy can be. A dataset on which no model rises meaningfully above chance cannot separate good architectures from bad ones: whatever ranking it produces is dominated by label noise. CMOSE is therefore the development dataset; DAiSEE re-enters as a transfer source and target in Section 4.5 and in Chapter 5, where its difficulty is informative rather than obstructive.

Table 4.2: Best in-domain result per dataset (CMOSE vs DAiSEE).

In-domain dataset	Best model	Loss	QWK	Macro-Acc	Macro-MAE	Accuracy
CMOSE	openface_tcn	CE	0.537	0.535	0.547	0.761
DAiSEE	i3d_mlp	CE	0.166	0.289	1.226	0.548

4.2.3 Development Loss

In-domain, the loss axis trades the primary metric against the balanced secondary ones. Figure 4.2 shows, for every baseline, that rebalancing the loss from cross-entropy through weighted cross-entropy to the ordinal loss raises macro-accuracy and improves macro-MAE while QWK falls for four of the five baselines: the balanced losses buy minority-class recall at the cost of chance-corrected agreement. The trade is Pareto: no loss is best on QWK and on the balanced secondary metrics at once. For the I3D MLP, for instance, switching from cross-entropy to the ordinal loss lifts macro-accuracy from 0.526 to 0.635

and improves macro-MAE from 0.585 to 0.458 (the best balanced figures of any baseline) while QWK falls from 0.519 to 0.487.

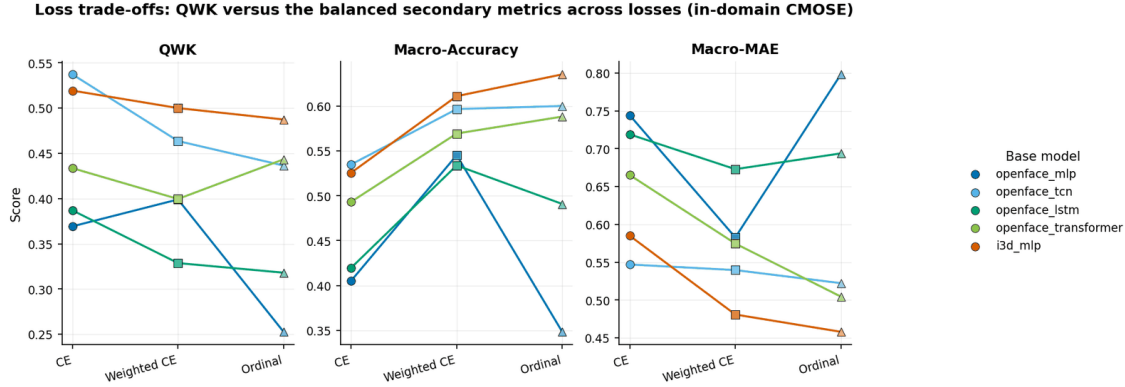


Figure 4.2: What each loss trades: QWK (the selection metric) against the balanced secondary metrics (macro-accuracy and macro-MAE) for every baseline across the three losses, in-domain on CMOSE. Macro-accuracy and macro-MAE improve as the loss is rebalanced, while QWK is generally highest under cross-entropy.

Protocol. The strongest ordinal agreement of the whole grid is obtained under cross-entropy (best-architecture QWK by loss: CE 0.537, weighted CE 0.500, ordinal 0.487), so cross-entropy is the development loss: it maximises QWK, the selection metric, and isolates the effect of the architecture from the confound of the loss. The hybrid of Chapter 5 is consequently trained with it. Deferred evidence: Section 4.5.2 shows this CE > weighted CE > ordinal ranking is preserved intact under domain shift, which after the fact makes cross-entropy the reliable choice, not merely the convenient one. The training and validation loss curves of every configuration converge stably and early-stop well before the epoch cap (Section 3.7), so the differences reported in this chapter are architectural rather than optimisation artefacts.

4.3 In-Domain Results

Five single-encoder baselines are trained and tested in-domain on CMOSE under each of the three losses (four on the OpenFace descriptor: an MLP, a TCN, an LSTM, and a Transformer; and one on the I3D clip vector, the I3D MLP); Figure 4.3 reports the complete grid on all six metrics. On QWK, the selection metric, the strongest baseline is the OpenFace TCN under cross-entropy at QWK 0.537 (max-over-loss, per Section 4.1): the reference level the proposed architecture must exceed.

The grid adds two structural observations. First, the loss axis determines on which metric family a model is strongest: cross-entropy is strongest on the micro and agreement metrics, while only the rebalanced losses recover the rare disengaged classes that macro averaging weights equally, a direct symptom of the

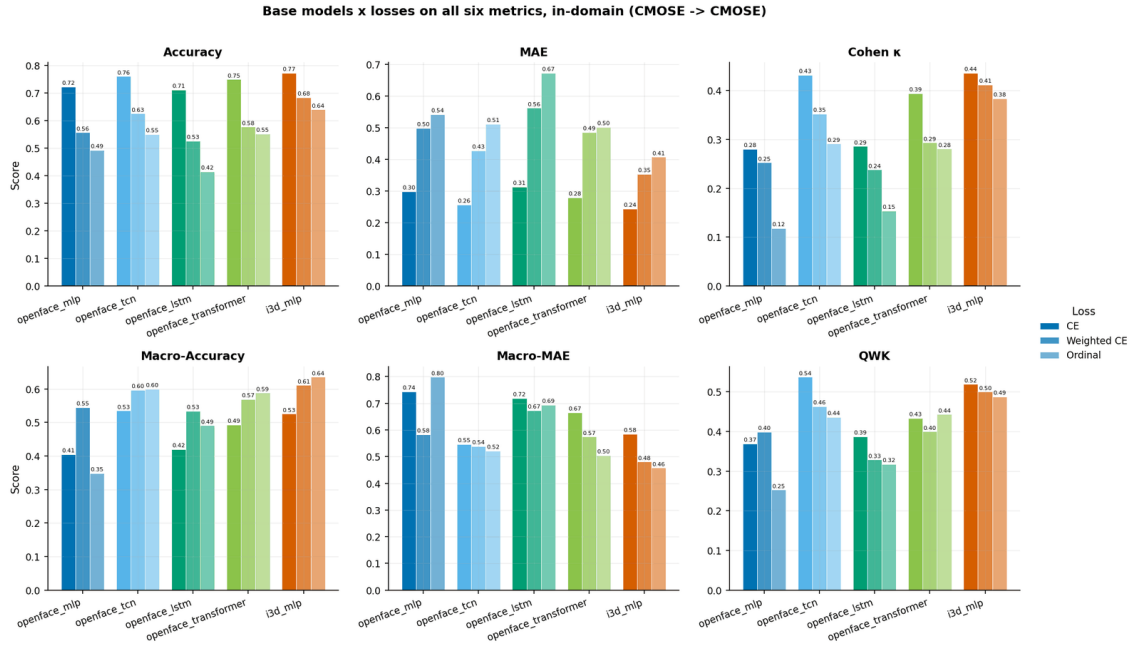


Figure 4.3: The five base models under the three losses on all six metrics, in-domain on CMOSE. The metrics disagree on the winner: accuracy, MAE, and Cohen’s kappa favour the I3D MLP under cross-entropy, QWK favours the OpenFace TCN, and the macro metrics peak under the balanced losses.

class imbalance. Second, the architecture axis determines which kind of feature is strongest: the OpenFace TCN’s per-frame facial dynamics carry the cues (a head turning away, gaze drifting off-screen) that separate the rarer disengaged levels and rank highest on QWK, while the I3D MLP, over a single pooled appearance vector, aligns with the majority Engage class and ranks highest on every micro metric. On QWK, the selection metric, the two score almost the same (0.537 versus 0.519). Two models this close, yet built on disjoint feature families, pose a question the score table cannot answer: are they interchangeable, or do they succeed on different clips?

4.4 Prediction Agreement

Scores summarise how often a model is right; they say nothing about which clips it is right on. This section descends to individual predictions on the 1,221 in-domain CMOSE test clips, first across the whole grid, then for the two feature families.

4.4.1 Agreement between Configurations

Figure 4.4 reports the pairwise Cohen’s kappa between the predictions of every pair of configurations. Mean pairwise agreement is only $\kappa = 0.27$ – 0.39 , far below what the clustered scores of Figure 4.3 suggest, and sharing the architecture ($\kappa = 0.391$) or the loss ($\kappa = 0.374$) raises it only mildly over sharing neither

($\kappa = 0.273$): the fifteen configurations are different hypotheses about the data, not reparametrisations of one another.

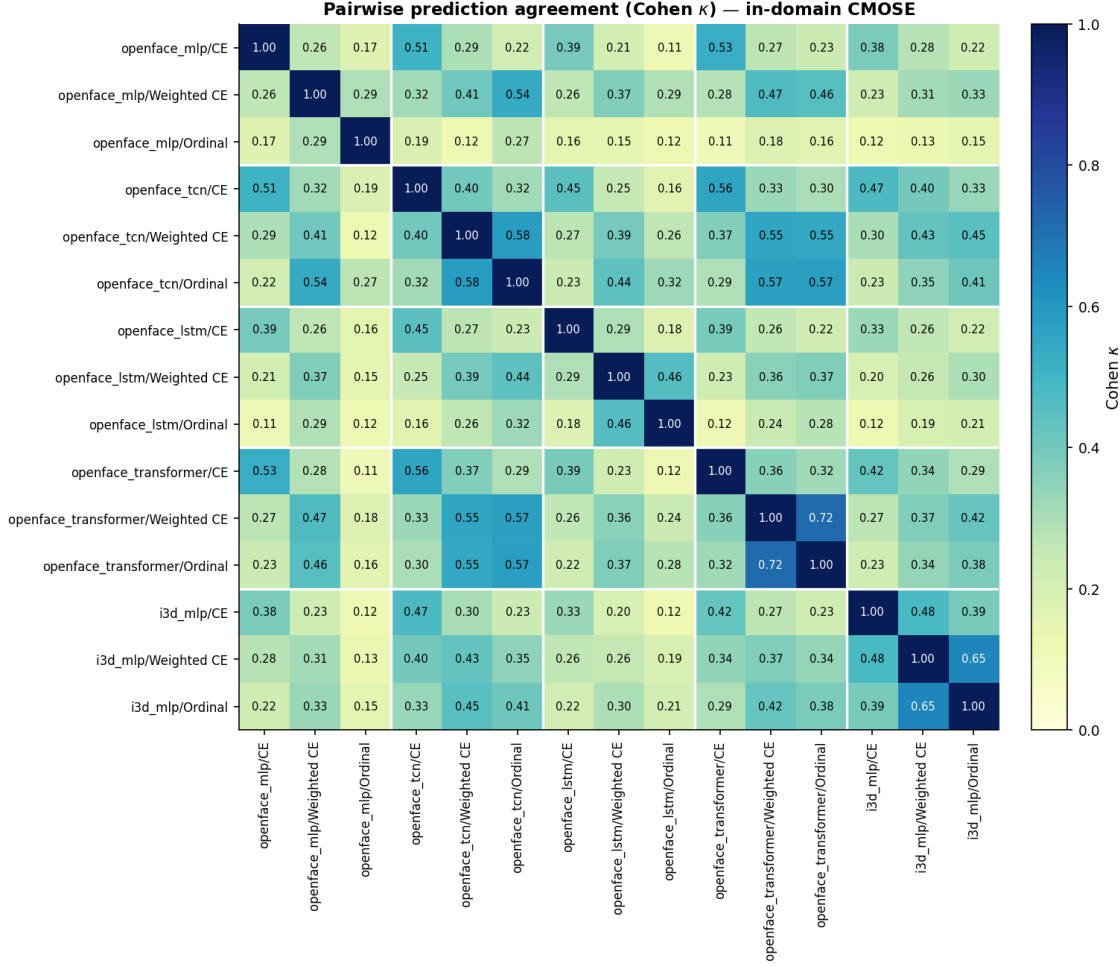


Figure 4.4: Pairwise prediction agreement (Cohen’s κ) between all fifteen baseline configurations on the in-domain CMOSE test set, in model-major blocks. The diagonal blocks are the same architecture under different losses; agreement is moderate at best.

The practical consequence is headroom. The best single configuration classifies 77.3% of the test clips correctly, but at least one of the fifteen is correct on 97.6% (the oracle); only 2.4% of clips defeat every model. Yet plain majority voting over the fifteen lands at 73.5%, below the best single model, because the errors are too decorrelated for unweighted voting to use. The headroom exists, but claiming it requires a learned combination of complementary views of the input. This is the design premise of the feature-group hybrid of Chapter 5, which fuses separately encoded behavioural streams instead of averaging finished predictions.

4.4.2 OpenFace versus I3D

The two families scored almost the same in Section 4.3, yet they are different in nature: OpenFace is a per-frame, structured account of facial behaviour (gaze, eye and face landmarks, head pose, action units), while I3D is a single clip-level

vector of global appearance and motion. Table 4.3 compares their best cross-entropy models: the OpenFace TCN is strongest on QWK and on both balanced secondary metrics, so by scores alone the I3D MLP appears merely redundant.

Table 4.3: Best OpenFace encoder (selected on QWK) vs the I3D MLP on QWK and the balanced secondary metrics (in-domain CMOSE, cross-entropy).

Feature family	Model	QWK	Macro-Acc	Macro-MAE
OpenFace (behaviour descriptors)	openface_tcn	0.537	0.535	0.547
I3D (appearance/motion)	i3d_mlp	0.519	0.526	0.585

The agreement map says otherwise. In Figure 4.4, the I3D MLP’s row is the palest in the grid: its agreement with the OpenFace TCN is among the lowest of any pair, despite the two having the closest scores. Two models that score alike but agree this rarely must succeed on different clips, and the clip-level decomposition confirms it: against the OpenFace TCN, the I3D MLP has the largest exclusive-correct share of any pair in the grid (8.6% of all test clips are clips only I3D gets right) and the smallest both-wrong share (15.3%). The pair’s oracle accuracy is 84.7%, +7.4 points over the best single model and the largest pair gain in the grid.

Conclusion. The behaviour descriptors and the appearance features are looking at different evidence: the families fail on different clips. This is the empirical motivation for fusing the two streams in the hybrid of Chapter 5, a motivation the score table alone could never supply.

4.5 Cross-Dataset Generalisation

In-domain numbers are not the deployment question. This section trains on one dataset and tests on another, with three aims: measure whether any ordinal signal is preserved under a change of dataset, close the two arguments the protocol deferred, and ask whether in-domain strength predicts generalisation at all.

4.5.1 The 3×3 Matrix

The collapse. The 3×3 heatmaps of Figure 4.5 are the central generalisation result for the baselines. On the diagonal (train and test on the same dataset) QWK is healthy (CMOSE 0.54), macro-accuracy is well above chance (CMOSE 0.64), and ordinal error is low (CMOSE macro-MAE 0.46). Off the diagonal all three collapse together: $\text{CMOSE} \rightarrow \text{DAiSEE QWK} = 0.02$ and $\text{DAiSEE} \rightarrow \text{CMOSE} = 0.05$ (chance-level agreement), macro-accuracy falls toward the 0.25 chance floor, and macro-MAE inflates toward a full class. The collapse is roughly symmetric (neither public dataset transfers to the other), which rules out one dataset being a superset of the other and points to a real shift in both label priors and recording conditions.

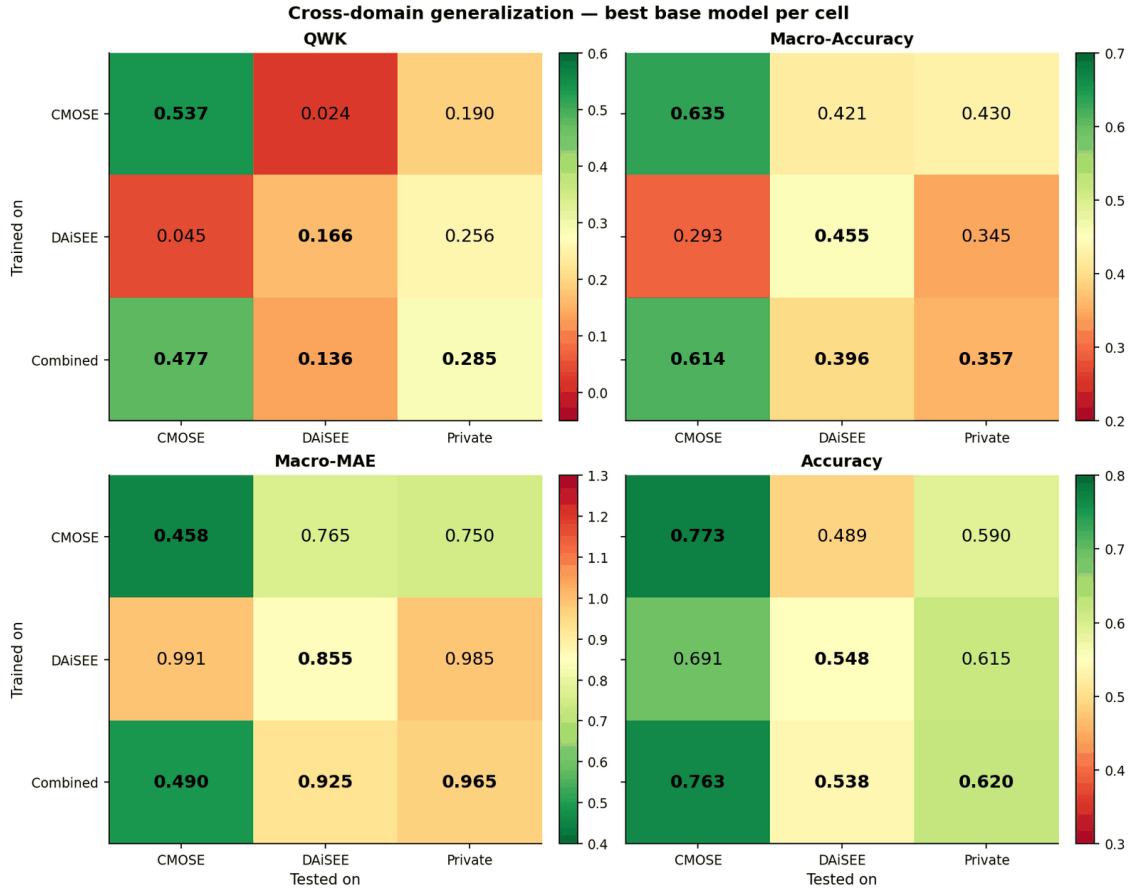


Figure 4.5: Cross-domain generalisation, best base model per cell (selected by QWK), on QWK (the selection metric) and the balanced secondary metrics macro-accuracy and macro-MAE, plus raw accuracy, the misleading one: accuracy stays high in exactly the off-diagonal cells where every chance-corrected metric collapses.

The architecture ranking is reshuffled. Comparing, per architecture, the in-domain QWK with the mean QWK over the unseen-target cells (the cells whose test dataset contributed no training data: the two cross-dataset cells and the private column), the in-domain order $\text{TCN } 0.537 > \text{I3D } 0.519 > \text{Transformer } 0.443 > \text{MLP } 0.399 > \text{LSTM } 0.387$ becomes $\text{TCN} \approx \text{Transformer} > \text{LSTM} > \text{I3D} > \text{MLP}$ on unseen targets. The I3D MLP falls from second place to fourth: its global appearance features bind to the dataset, while the OpenFace behaviour descriptors carry more signal that holds across datasets.

The fix: combine the source datasets. The most useful cell is the unseen Private column: training on the Combined dataset gives the best private-set QWK (0.285), exceeding DAiSEE-only (0.256) and CMOSE-only (0.190), while keeping near-in-domain CMOSE performance (0.477, the Combined→CMOSE cell of Figure 4.5). The combination of CMOSE and DAiSEE is more diverse than either dataset alone, covering a wider range of label priors and recording conditions, so a model trained on this training set generalises better and transfers better to an

unseen target such as the private set. This is the most effective design choice for an unknown target, and Chapter 5 combines it with the architecture contribution on the decisive private set (RQ4).

4.5.2 Closing the Loop: Metric and Loss Reliability

Accuracy cannot see the collapse (completes Section 4.2.1). In the accuracy panel of Figure 4.5, CMOSE→DAiSEE shows a deceptively reasonable 0.49 and DAiSEE→CMOSE 0.69, in the same cells where the QWK panel reads 0.02 and 0.05, because predicting near the majority class scores well on any imbalanced target. A metric that appears normal while the ordinal signal is gone is misleading rather than reassuring: accuracy is the least reliable metric here and QWK (the only metric that bridges both blocks, and the one that detects the collapse) the most.

The loss ranking is preserved under the shift (completes Section 4.2.3). On the unseen-target cells the best-architecture QWK by loss follows the same $\text{CE} > \text{weighted CE} > \text{ordinal order}$ as in-domain. The loss is the only factor in the grid whose ranking transfers (the architecture ranking reshuffles): choosing cross-entropy in-domain remains the right choice out of domain, which confirms, after the fact, the choice to establish it as the development loss.

4.5.3 In-Domain versus Transfer

Does in-domain strength predict transfer? Pairing, for each of the fifteen baseline configurations, its in-domain QWK (trained and tested on CMOSE) with its deployment transfer QWK (trained on the Combined dataset and tested on the held-out private set) gives one point per configuration on a single scatter. The Spearman rank correlation is only $\rho = 0.31$ ($p = 0.25$, not significant), so a strong in-domain score is at best a weak signal of private-set transfer and the in-domain leaderboard is not a reliable basis for selecting a model for an unknown target; selection must instead rely on combining datasets. Chapter 5 re-examines the same relationship on the 486-configuration hybrid population, where it stays weak ($\rho = 0.27$, Figure 5.6): the architecture lifts both in-domain and transfer QWK without making one a reliable predictor of the other.

4.6 Chapter Summary

The evaluation protocol established QWK as the single primary metric that selects the best model (with macro-accuracy, macro-MAE, accuracy, micro-MAE, and Cohen’s kappa kept as secondary, reported but never selecting), CMOSE as the development dataset, and cross-entropy as the development loss. Under it, the strongest baseline is the OpenFace TCN at QWK 0.537, the reference level the hybrid must exceed. The prediction-level analysis showed the configurations agree far less

than their scores suggest (oracle 97.6%) and that the OpenFace and I3D families fail on different clips, the direct motivation for fusing them. The cross-dataset study showed baselines do not transfer to a new dataset, confirmed the metric and loss choices, and made dataset combination the most effective simple fix. Chapter 5 now asks whether learning to fuse the complementary streams outperforms these baselines, and whether the architecture gain and dataset combination reinforce each other on unseen data.

CHAPTER 5. FEATURE-GROUP HYBRID

5.1 Overview

Chapter 4 established the evaluation frame (QWK as the single primary metric that selects the best model, with macro-accuracy and macro-MAE among the secondary metrics; CMOSE as development dataset; cross-entropy as development loss; the OpenFace TCN at QWK 0.537 as the reference level) and supplied the design motivation: the baseline configurations, and the OpenFace and I3D families in particular, fail on different clips. This chapter evaluates the proposed feature-group hybrid (Section 3.5) against that frame: all $3^5 = 243$ per-group encoder configurations, each with and without the I3D stream, trained with cross-entropy. Section 5.2 first re-validates the Chapter 4 frame on this 486-configuration population (RQ1 in part); Section 5.3 isolates which behavioural group’s encoder choice matters (RQ2); Section 5.4 measures the effect of I3D fusion with a paired ablation (RQ3); Section 5.5 pins down the headline in-domain comparison (RQ1); and Section 5.6 takes the hybrid out of domain, establishing that it generalises better than the baselines in every cell of the matrix, and most of all on the decisive private set I collected, where the architecture and dataset combination reinforce each other (RQ4). Section 5.7 discusses why. As in Chapter 4, family-level claims are reported as means and distributions and headline comparisons as the tuned maximum of QWK, the single selection metric (Section 4.1).

5.2 Consistency with the Baseline Frame

The ablation population also serves as a robustness check of Chapter 4’s protocol: 486 configurations of a different architecture family, none of which existed when the frame was established. Figure 5.1 plots the distribution of the two hybrid families (with and without I3D) on every metric, each panel against the best base model on that metric. Three findings of Chapter 4 reproduce.

The metric structure reproduces. Recomputing the Kendall rank correlation between the six metrics over the 486 in-domain configurations yields the same two-block picture as Section 4.2.1: a micro block, a macro block, weak cross-block agreement, and QWK again the only bridge between them, which reflects the metrics on this task, not any model family.

Accuracy remains insensitive. In Figure 5.1 every configuration falls within a narrow accuracy range around the QWK-selected baseline’s 0.761 (top-left panel), so a model chosen on accuracy would see no architecture signal at all, while the QWK panel separates the families cleanly. Because the reference line is the same

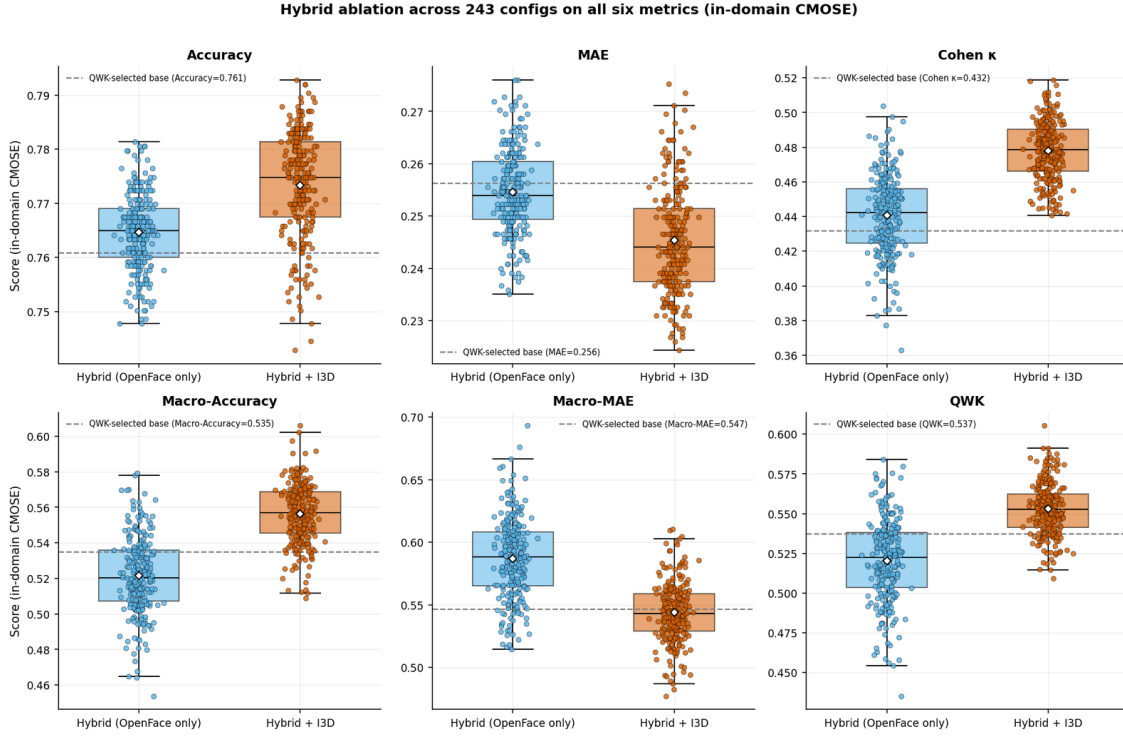


Figure 5.1: Hybrid ablation across all 243 per-group encoder configurations, with and without I3D, on all six metrics (in-domain CMOSE). The dashed line in every panel is the same model (the single QWK-selected baseline: OpenFace TCN, cross-entropy) read on that panel’s metric, not a per-metric champion.

QWK-selected baseline in every panel, the macro panels show the hybrid family clearing that one model on balanced recall and ordinal error too (macro-accuracy and macro-MAE), rather than merely matching a different per-metric champion.

The cross-domain collapse reproduces. Taken out of domain (Section 5.6.1), the hybrid matrix shows the same shape as the baseline matrix: a healthy diagonal (0.605 on CMOSE, 0.228 on DAiSEE) and an off-diagonal collapse, with accuracy again masking it. The collapse comes from the nature of the problem, divergent label priors and recording conditions, not from a particular architecture.

All three pillars of the Chapter 4 frame thus hold across a 32-fold larger model population; the protocol carries over without amendment (RQ1’s frame condition).

5.3 Encoder Choice per Group

The 243-configuration grid varies five group→encoder choices at once; this section marginalises them to find which group’s choice actually moves QWK, the selection metric (RQ2), and whether any preference is specific to one dataset or is preserved under shift.

Table 5.1 reports the marginal in both regimes. In-domain, only head pose shows a clear preference: TCN (mean QWK 0.545), ahead of LSTM (0.541) and

Transformer (0.523), a spread of 0.022; macro-MAE agrees, lowest (best) under the TCN. The other four groups are insensitive to the encoder (QWK spread ≤ 0.006). Under shift (the same marginal over the unseen-target cells), head pose again has the largest spread (0.010) and again prefers the TCN, while the other groups' spreads (0.002–0.004) flip winners arbitrarily, which is just noise.

Table 5.1: Per-group marginal effect of encoder choice on mean QWK, in-domain (CMOSE) and pooled over the unseen-target cells.

Group	Regime	TCN	Transformer	LSTM	Spread	Prefers
Gaze	In-domain	0.539	0.537	0.534	0.006	TCN
Gaze	Unseen targets	0.093	0.095	0.094	0.002	Transformer
Eye landmarks	In-domain	0.538	0.537	0.535	0.003	TCN
Eye landmarks	Unseen targets	0.096	0.092	0.094	0.004	TCN
Face landmarks	In-domain	0.537	0.534	0.539	0.006	LSTM
Face landmarks	Unseen targets	0.092	0.096	0.093	0.004	Transformer
Head pose	In-domain	0.545	0.523	0.541	0.022	TCN
Head pose	Unseen targets	0.100	0.091	0.090	0.010	TCN
Action units	In-domain	0.536	0.537	0.537	0.001	Transformer
Action units	Unseen targets	0.093	0.096	0.093	0.003	Transformer

The consistency across regimes shows the head-pose rule is not a tuning accident but reflects the signal itself: head dynamics such as nodding and turning away are temporally local and suit convolution (RQ2). The design space is otherwise flat, so most of the architecture's benefit must come from decomposing and fusing rather than from the per-group encoder search, which is what the next section measures.

5.4 Effect of I3D Fusion

Chapter 4 predicted, from the error-overlap analysis of Section 4.4.2, that fusing the I3D stream should help. This section measures the prediction with the strongest design in the grid: pairing each OpenFace-only configuration with its exact +I3D twin (same five encoders), so the distribution of paired QWK differences is the I3D effect, free of the configuration as a confounder, in three regimes.

Where the test dataset is seen in training, fusion helps. Over the seen-target cells the mean paired gain is +0.060 QWK, and 94% of the 972 pairs improve (Table 5.2). In-domain on CMOSE the complementarity of Section 4.4.2 produces a clear gain: the I3D-fused family is centred above the OpenFace-only family (median QWK 0.553 against 0.522, visible in Figure 5.1), and 82% of the 243 fused configurations exceed the 0.537 baseline reference, against 26% of the OpenFace-only ones. The gain belongs to the whole family, not to a single favourable configuration (RQ3).

Table 5.2: Paired effect of adding the I3D stream on QWK, by evaluation regime (each OpenFace-only hybrid against its exact +I3D twin).

Regime	Pairs (n)	Mean Δ QWK	Pairs improving (%)
Seen target	972	+0.060	94
Cross-corpus (public)	486	-0.021	26
Private (unseen)	729	+0.005	52

Beyond the training datasets, the gain evaporates. Cross-dataset the mean paired effect is -0.021 QWK with only 26% of pairs improving; on the private set it is $+0.005$ with 52% improving and high variance across configurations (Table 5.2). I3D’s global appearance features carry dataset bias along, the same mechanism behind the I3D MLP’s fall from second to fourth in the ranking under shift (Section 4.5.1). The rule (RQ3): fuse I3D when the target domain is represented in training; rely on the OpenFace groups otherwise, a rule about the family average (the best single private-set model does carry I3D, but it is one tail of a wide, zero-centred distribution).

5.5 Comparison with Baselines

Best hybrid versus best baseline. Table 5.3 lists the top-5 configurations; the best overall is the I3D-fused hybrid TCN_T_TCN_LSTM_T, and every top configuration mixes at least two encoder types. On QWK, the single selection metric, the best hybrid improves on the best baseline by $+0.068$ (0.605 versus 0.537), and the secondary balanced metrics corroborate it (macro-accuracy 0.582 versus 0.535, macro-MAE 0.489 versus 0.547). The selection metric and the secondary metrics move in the same direction, so the gain is real but modest. It is the CMOSE diagonal cell of the cell-by-cell comparison in Section 5.6.1 (Figure 5.3).

Table 5.3: Top-5 hybrid configs in-domain (CMOSE), sorted by QWK.

Variant	Arch (gaze_eye_face_head_au)	QWK	Macro-Acc	Macro-MAE	Accuracy
Hybrid + I3D	TCN_T_TCN_LSTM_T	0.605	0.582	0.489	0.784
Hybrid + I3D	T_TCN_TCN_LSTM_T	0.591	0.606	0.483	0.787
Hybrid + I3D	T_T_LSTM_LSTM_TCN	0.591	0.563	0.505	0.781
Hybrid + I3D	T_TCN_T_TCN_T	0.588	0.566	0.531	0.789
Hybrid + I3D	LSTM_LSTM_TCN_LSTM_T	0.588	0.567	0.517	0.787

Where does the gain land? The per-class scores localise it. Relative to the best baseline, the hybrid improves recall on the ordinal extreme Highly-Engage ($0.38 \rightarrow 0.51$) and on Disengage ($0.45 \rightarrow 0.55$), while keeping Engage near 0.90. The rarest class Highly-Disengage, however, stays unsolved and is slightly traded

away (recall $0.40 \rightarrow 0.37$): the model reallocates capacity toward the more frequent extreme. The per-class figure is reserved for the decisive private set (Figure 5.5, Section 5.6.2), where the same uneven signature recurs in sharper form.

The in-domain improvement is therefore real but uneven: the hybrid clears the baseline on QWK and on the balanced secondary metrics by strengthening the well-populated upper-engagement region and the middle disengaged class, while the $\sim 2\text{--}3\%$ -frequency Highly-Disengage class remains the main open challenge (RQ1). The next section shows the picture becomes more severe still out of domain, and that the architecture’s advantage over the baseline widens there.

5.6 Cross-Dataset Generalisation

5.6.1 Cross-Dataset Matrix

The hybrid cross-dataset heatmap (Figure 5.2) shows the same pattern as the baseline matrix of Section 4.5: a healthy diagonal and an off-diagonal collapse of QWK and the balanced secondary metrics alike: the hybrid does not rescue cross-dataset transfer. The misleading behaviour of accuracy persists: on DAiSEE $\rightarrow$ CMOSE the best QWK any of the 486 hybrid runs achieves is 0.10, yet hybrids in that cell reach accuracies up to 0.69. A second architecture family thus lands in the same place (the metric frame is not an artefact of the single-encoder models), and this is the last time accuracy appears in the thesis’s analysis.

Within the collapse, the hybrid is strongest in every cell. Figure 5.3 subtracts the baseline matrix (Figure 4.5) from the hybrid matrix (Figure 5.2) cell by cell: the best hybrid outperforms the best baseline in all nine train $\times$ test cells, with every cell of the Δ QWK matrix positive. The gains are smallest on the in-domain diagonal ($+0.068$ on CMOSE) and largest on the decisive Private column ($+0.118$, $+0.050$, $+0.094$ from the three training sources). The improvement is not a single tuned point but a uniform shift of the entire matrix, the strongest form the family-level claim can take. Read column by column, the figure ties the two headline comparisons together: its CMOSE diagonal cell is the in-domain gain of Section 5.5, and its Private column is the private-set result developed next.

5.6.2 The Private Set

The main generalisation test in this thesis is the private set: the 366 clips I collected and hand-labelled myself (Section 3.2.2), processed through the identical OpenFace/I3D pipeline and used for testing only. Because no model is ever trained on it, it is a test-only, in-the-wild set, with its own recording conditions and label prior (58% Engage, 31% Highly-Engage, only 2.7% Highly-Disengage). The only design lever is the choice of training source. The Private column of Figure 5.3 and

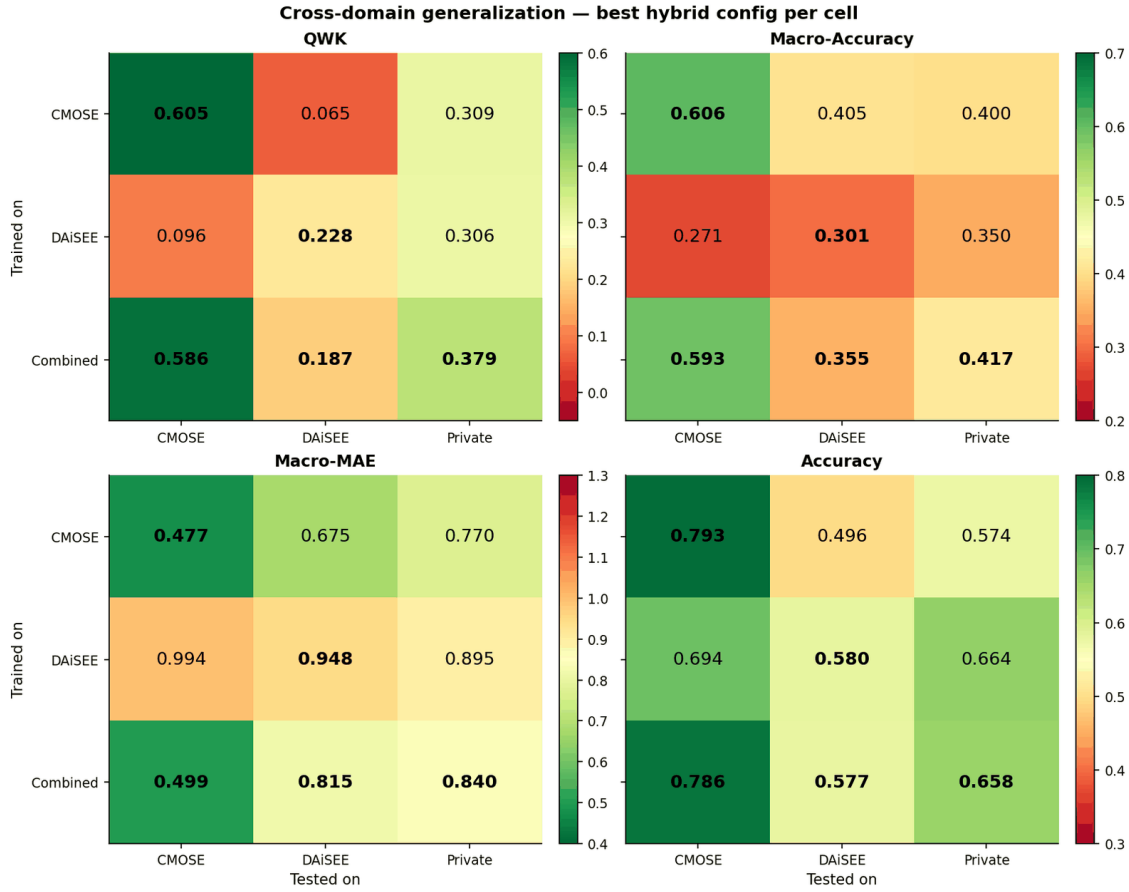


Figure 5.2: Cross-domain generalisation, best hybrid per cell (selected by QWK), on QWK (the selection metric) and the balanced secondary metrics macro-accuracy and macro-MAE, plus raw accuracy, which again stays high in the off-diagonal cells where the chance-corrected metrics collapse.

Table 5.4 report the best base model and the best hybrid per training source, and two effects combine.

Table 5.4: Private set (test-only): best base vs best hybrid by training source.

Train source	Best base (model/loss)	Base QWK	Base macro-MAE	Best hybrid (arch)	Hybrid QWK	Hybrid macro-acc	Hybrid macro-MAE
CMOSE	openface_transformer/ce	0.190	0.750	Hybrid (OpenFace only) TCN_T_LSTM_TCIN_T	0.309	0.385	0.914
DAiSEE	i3d_mlp/ce	0.256	1.031	Hybrid + I3D LSTM_LSTM_TCIN_LSTM_LSTM	0.306	0.350	0.966
Combined	openface_transformer/ce	0.285	0.997	Hybrid + I3D T_T_T_LSTM_T	0.379	0.385	0.877

- (i) **Training source matters, and the combined dataset performs best.** For both model families, training on Combined generalises best to the private set (best base QWK: Combined 0.285 versus DAiSEE 0.256 versus CMOSE 0.190): combining mixed source datasets is the most effective simple route to an unseen target, exactly as Chapter 4 predicted.
- (ii) **The hybrid generalises better, and helps more here than in-domain.** The hybrid outperforms the best base model on QWK for every training source, and the gap is largest exactly where it matters: under combined training, QWK 0.379

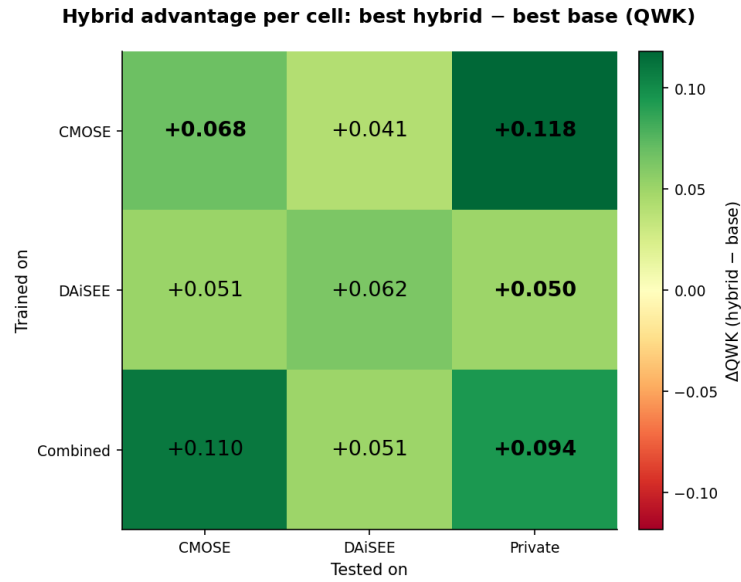


Figure 5.3: Cell-by-cell hybrid advantage: best-hybrid minus best-base QWK over the 3×3 train $\times$ test matrix (Figure 5.2 minus Figure 4.5). Every cell is positive (the hybrid is strongest across the whole matrix), and the gain is largest on the unseen, self-collected Private column. The in-domain CMOSE cell and the Private column (bold) are the two comparisons read in Sections 5.5 and 5.6.2.

versus 0.285, a +0.094 improvement, wider than the +0.068 in-domain gain, so distribution shift amplifies rather than erodes the architecture’s advantage. The best private-set model overall (selected, as everywhere, on QWK alone) is the combined-trained I3D-fused hybrid `T_T_T_LSTM_T` at QWK 0.379; the secondary macro-MAE agrees, at the lowest value of any model (0.877 against the best base’s 0.997). (One caveat in Table 5.4: the CMOSE-only base model shows a deceptively low macro-MAE (0.75) despite the weakest QWK by predicting conservatively around the majority classes, exactly the misleading reassurance that selecting on QWK alone, rather than on a balanced-error metric, is designed to avoid; the secondary metrics are reported to expose such cases, never to reward them.)

Figure 5.4 compares the confusion of the two combined-trained models on the private set. The hybrid captures the ordinal structure better: Engage recall rises from 0.46 to 0.72 and errors concentrate on the diagonal and its neighbours, where the base model scatters Engage mass into Highly-Engage (0.44). As in-domain, the rare disengaged classes are the failure mode: hybrid Disengage recall is only 0.29, and Highly-Disengage collapses to 0.00 (all ten private HD clips are misread).

Where the private-set gain lands. The per-class F1 scores (Figure 5.5) localise the +0.094 QWK gain, and the signature is the same as in-domain (Section 5.5) but sharper. The hybrid lifts F1 on the well-populated Engage (0.52 $\rightarrow$ 0.70) and

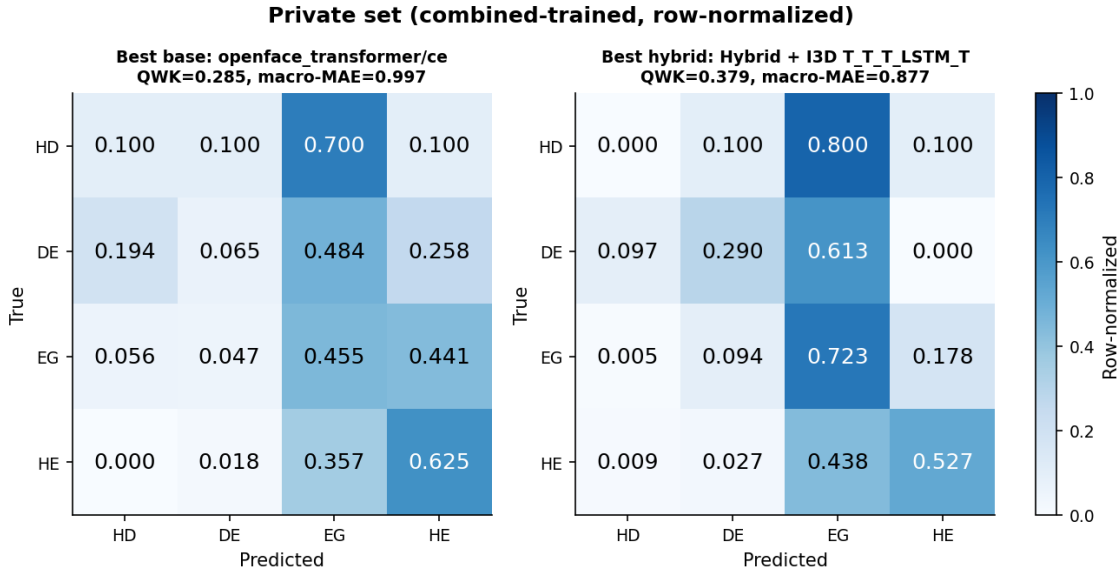


Figure 5.4: Private-set confusion of the combined-trained best base model versus best hybrid (row-normalised).

Highly-Engage ($0.49 \rightarrow 0.56$) classes, and, where the base model is all but flat, gives the middle Disengage class real mass ($0.09 \rightarrow 0.28$). The rarest Highly-Disengage class, by contrast, is traded all the way to zero ($0.07 \rightarrow 0.00$): out of domain the model concentrates its limited capacity on the engaged region even more sharply, and the rare disengaged extreme remains the thesis’s open failure mode (RQ4).

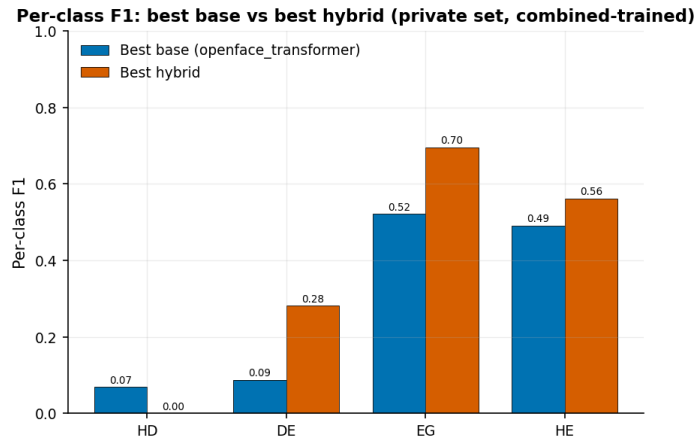


Figure 5.5: Per-class F1 on the private set: combined-trained best base model versus best hybrid. The hybrid’s out-of-domain gain concentrates on the engaged region and the middle Disengage class, while Highly-Disengage collapses to zero.

5.6.3 In-Domain versus Transfer

Section 4.5.3 asked, for the baselines, whether in-domain QWK predicts transfer, pairing each configuration’s in-domain cell (trained and tested on CMOSE) with its deployment cell (trained on Combined and tested on the held-out private set). Figure 5.6 repeats that scatter and overlays the 486 hybrid configurations; each

configuration is a single point pairing its own two runs, so the axes pool no training populations. Across the hybrid configurations the correlation stays weak ($\rho = 0.27$, significant only because $n = 486$), in the same range as the baseline relationship ($\rho = 0.31$, $n = 15$): a configuration’s pure in-domain CMOSE ability is, at most, a weak predictor of how it transfers to the private set. The architecture’s contribution therefore shows as position rather than slope: the hybrid cloud lies above and to the right of the baseline configurations, reaching both higher in-domain QWK and higher private-set transfer, so the feature-group design lifts both ends together, while combined training remains the lever that places a model in that upper-right region. The private-set winner is thus found by family-level robustness and combined training, not by reading the in-domain leaderboard.

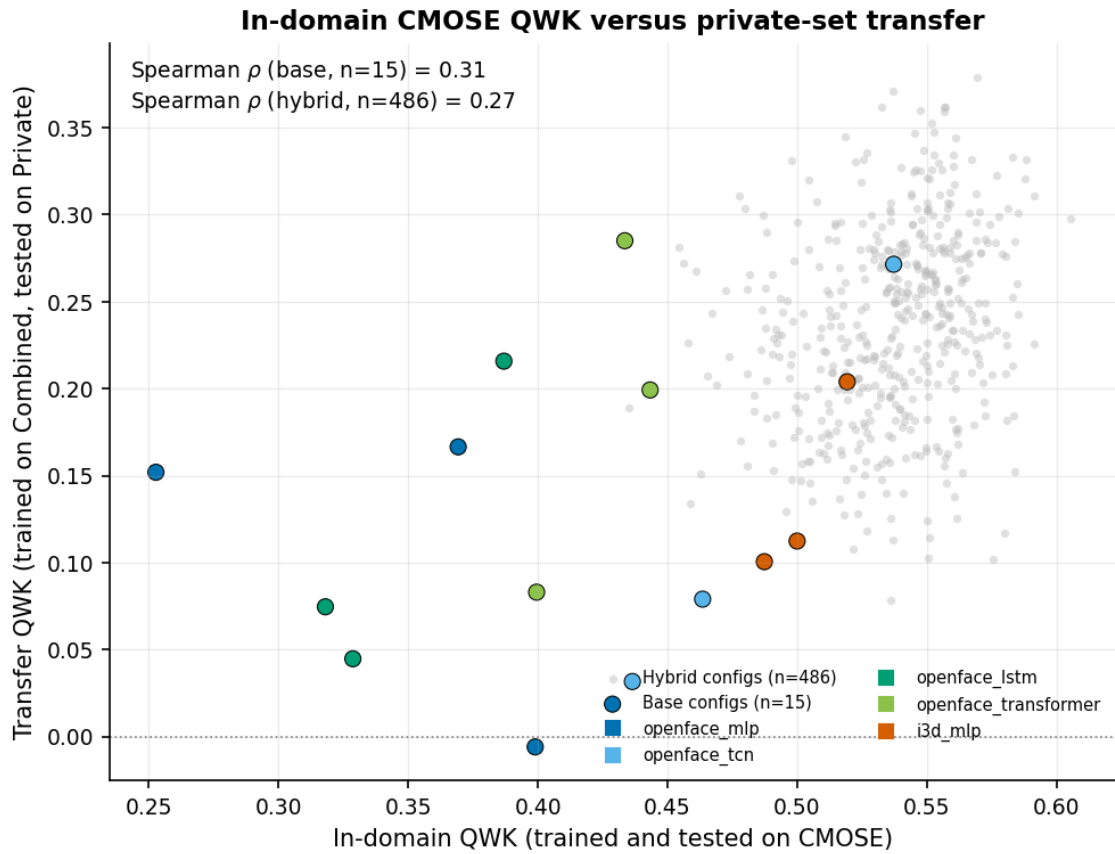


Figure 5.6: In-domain QWK (trained and tested on CMOSE) versus transfer QWK (trained on Combined, tested on the held-out private set); each configuration is one point pairing its own two runs, so the axes pool no training populations. The correlation is weak for both the 486 hybrid configurations ($\rho = 0.27$) and the 15 baselines ($\rho = 0.31$). The hybrid cloud lies above and to the right of the baselines: the architecture improves both in-domain QWK and private-set transfer.

5.7 Discussion

Why the levers work. Combining datasets works through prior coverage: CMOSE and DAiSEE are imbalanced in opposite directions (Section 3.2), so

their combination spans a wider range of label priors and an unseen target is closer to the combined dataset than to either single source. The hybrid’s margin widens off-domain for the reason it was designed: each behavioural channel is encoded separately and forced to be discriminative through its auxiliary head, so that the representation is less dependent on the distinctive features of any dataset. The two feature families diverge under shift because the OpenFace descriptors live in subject-centred physical units that mean the same thing in any setup, whereas the I3D embedding encodes global appearance, what changes between datasets, so fusion adds signal only where the test statistics are represented in training.

What the headline number means. QWK 0.379 is a clear improvement over chance, every single-source model, and the best baseline (+0.094), but in absolute terms moderate: with macro-MAE 0.877 the model captures the broad engaged-versus-disengaged trend yet is unreliable at fine distinctions, and the rarest class Highly-Disengage fails in both regimes. It is usable as an aggregate, classroom-level signal but not to flag an individual disengaged learner without human confirmation. The open problem is one of data and objective (no encoder assignment can recover a class the training distribution barely contains), which is why the remedies for imbalance in Chapter 6 are the natural next step.

5.8 Chapter Summary

The Chapter 4 frame held under the 486-configuration robustness check: the block structure of the metrics, the insensitivity of accuracy, and the cross-domain collapse all reproduced as task properties. Only head pose has a clear encoder preference (a TCN), so the benefit comes from decomposing and fusing rather than the encoder search (RQ2). I3D fusion is positive where the test dataset is seen in training, neutral-to-harmful beyond it (RQ3). In-domain the best hybrid outperforms the best baseline on QWK, the single selection metric (0.605 versus 0.537), with the secondary metrics agreeing (RQ1); out of domain it is strongest in every cell of the matrix, and reaches the best private-set model when trained on the combined dataset (QWK 0.379, +0.094 over the baseline, wider than in-domain), and transfers better than the baselines at every in-domain level (RQ4), with Highly-Disengage the main failure mode in both regimes. Chapter 6 draws the findings together, revisits the research questions, and outlines future work.

CHAPTER 6. CONCLUSIONS

6.1 Summary

This thesis studied ordinal recognition of student engagement on a four-level scale, using the features OpenFace extracts from each frame and the features I3D extracts from the whole clip. It set out to improve recognition over standard single-encoder models and to describe the problem accurately, in particular its strong, uneven class imbalance and its behaviour under distribution shift.

For the method, the thesis built a complete and reproducible pipeline: feature extraction and leak-free preprocessing; a feature-group hybrid that splits the 709-dimensional OpenFace features into five behavioural groups (gaze, eye landmarks, face landmarks, head pose, action units), encodes each group with its own temporal encoder, and optionally adds a global I3D stream while training every stream with a small head; five single-encoder baselines; three loss functions; and a 3×3 cross-dataset evaluation protocol. The central idea was tested by training and evaluating all $3^5 = 243$ per-group encoder choices, with and without I3D, and by a self-collected, test-only private set of 366 hand-labelled clips that serves as the deciding deployment test.

The evidence supports five conclusions.

- (i) **The I3D-fused hybrid is the best in-domain model and, as a family, outperforms the strongest single-encoder baseline.** The I3D-fused family is centred above the baseline QWK of 0.537 (median 0.553, with 82% of its 243 configurations clearing the baseline), and the best configuration reaches QWK 0.605 with a mixed TCN/Transformer/LSTM assignment (TCN_T_TCN_LSTM_T). The gain over the best baseline is real but small (+0.068 QWK). The value lies in the framework and the full evidence, not in a jump in state of the art.
- (ii) **Most groups are not affected by which encoder they get; only head pose shows a clear preference, for a TCN** (a 0.022 QWK spread across the three encoders, against ≤ 0.006 for the other four groups). This is a simple design rule that reflects the short, local nature of head motion.
- (iii) **Engagement models do not generalise across datasets.** Off-diagonal QWK drops to near zero (CMOSE→DAiSEE 0.02) even when accuracy appears acceptable, so ordinal agreement metrics, not accuracy, must be reported on imbalanced engagement data.

- (iv) **On the self-collected, hand-labelled private set (test-only, in-the-wild), the three contributions combine.** The hybrid’s advantage is not limited to one comparison: it outperforms the best baseline in all nine train×test cells of the cross-dataset matrix. Because the private set is test-only, the only choice left is the training data; training on both datasets together with the feature-group hybrid gives the best result (QWK 0.379), outperforming the best baseline by +0.094 (against +0.068 in-domain) and surpassing every single-dataset hybrid. This is the thesis’s main practical result.
- (v) **The rare Highly-Disengage class is still the main open problem.** It is around 2–3% of the data, is traded away rather than solved by the hybrid in-domain, and drops to zero recall under distribution shift.

A note on scope: the architecture ablation is based on CMOSE because DAiSEE’s in-domain ordinal signal is too weak (QWK 0.166) to tell architecture differences apart from label noise. CMOSE is therefore used for development and the private set for the deciding check.

6.2 Synthesis

Read together, the two results chapters tell a single story whose lesson is not the one a benchmark paper usually reports. The thesis makes two kinds of contribution: it builds a feature-group hybrid and a strategy for combining datasets, and it gives a careful account of the problem’s imbalance, non-transfer, and the metrics that show them. The key finding is how the two work together.

The architecture matters more off-domain than in-domain. In-domain the feature-group decomposition is a small but clear improvement: the I3D-fused family scores higher than the baseline but the best configuration exceeds it by only +0.068 QWK, and most of the 243-cell design space is not affected by which encoder it gets. On its own, that is a small result. What makes it worth building is that the same decomposition gives the biggest gain exactly where it is hardest: on the unseen private set the gap over the baseline grows to +0.094, and it adds to dataset combination rather than fighting it. So the contribution is less a new in-domain state of the art than a representation that breaks down more slowly under distribution shift, which, for a problem whose only test is deployment, is what matters.

The fusion was measured before it was built. The choice to combine the OpenFace and I3D families did not rest on a hunch: the clip-level agreement analysis of Chapter 4 showed that the best OpenFace model and the I3D baseline disagree more with each other than OpenFace models do among themselves, and that each family gets clips right that the other cannot, evidence of different errors that the

paired I3D ablation of Chapter 5 then confirmed in-domain. The same analysis also measured unused room to improve: a perfect picker over the fifteen baseline configurations would reach 97.6% of clips against 77.3% for the best single model, which both motivates fusion and marks ensemble selection as a direction the thesis left aside on purpose.

The flat design space is itself the message. Four of five behavioural groups are not affected by their encoder, and only head pose prefers a TCN. This marks the limit of the contribution: the gain comes from splitting and fusing the features, not from a clever search over per-group encoders. This is a more transferable design rule than a single tuned configuration, and it is why the family-level effect, rather than any one cell, is reported as the result.

The negative results are contributions, not side notes. The near-total collapse of cross-dataset transfer, the way plain accuracy hides it, and the steady failure on the rarest disengaged class are not minor limits of these particular models. They are aspects of the problem that the field leaves unmeasured by staying in-domain and reporting accuracy. Showing them with numbers, and showing that QWK, macro-accuracy, and macro-MAE make them visible, changes engagement recognition from “beat the benchmark” to “report ordinal agreement and watch rare-class failure,” which is the practice the rest of this chapter recommends.

6.3 Revisiting the Research Questions

The four research questions posed in Section 1.3 can now be answered directly.

RQ1 (decomposition). Yes, decomposition helps. Encoding the five behavioural groups separately and fusing them outperforms encoding all 709 features with a single encoder: as a family the hybrid scores higher than the best baseline, and the effect holds across many configurations rather than arising from a single favourable configuration, with most of the I3D-fused configurations exceeding the baseline.

RQ2 (encoder assignment). Only head pose matters. Four of the five groups are basically not affected by whether their encoder is a TCN, a Transformer, or an LSTM (spread ≤ 0.006 QWK); head pose alone shows a clear preference for a TCN. The practical point is that the design space is flat and that a TCN is a safe default for every group, with head pose the one place the choice is worth making on purpose.

RQ3 (motion fusion). Yes in-domain, with a caveat under shift. When the model is tested on data like its training data, adding the clip-level I3D embedding helps: the paired comparison over the seen-target cells shows a mean gain of $+0.060$ QWK with 94% of pairs improving, and the fused family gives both the best in-domain

model and the best private-set model. Under cross-dataset shift, however, the same stream is neutral or slightly harmful on average (mean -0.021 on cross-public cells, $+0.005$ on the private set): the I3D stream carries information that is useful but specific to each dataset. So the fusion is recommended, but its gain should be claimed only for the setting where it holds.

RQ4 (generalisation and losses). Generalisation is poor and the metric choice is the key. Models do not transfer across datasets; off-diagonal ordinal agreement collapses while accuracy stays high and misleading. The in-domain score is at best a weak predictor of transfer: pairing each configuration’s CMOSE in-domain QWK with its Combined-trained private-set QWK gives only $\rho = 0.31$ for the baselines and $\rho = 0.27$ for the hybrids, so a model cannot be picked for deployment from its in-domain score alone. Among losses there is no overall winner (cross-entropy gives the best ordinal agreement, QWK, while weighted and ordinal losses recover minority recall and balanced ordinal error), so the loss is a trade-off. The cross-entropy-first order is the one in-domain preference that stays the same on every unseen target, which confirms, after the fact, the choice to establish it before development. QWK together with macro-accuracy and macro-MAE must be the main measure, and training on the combined datasets is the most useful tool for an unknown target.

6.4 Limitations

The findings should be read against several limits, each of which also motivates the future work below.

- (i) **Single-annotator private labels.** The private set was labelled by one annotator (myself) under the public-dataset rubric. Although the model suggestions shown during labelling were optional, the lack of a second annotator means no agreement between annotators can be reported for this set, and the 366-clip size limits how reliable the private-set conclusions are, especially for the ten Highly-Disengage clips, where a single wrong clip strongly affects the recall.
- (ii) **Clip-level I3D features.** The I3D features are a single pooled 1024-dimensional vector per clip (shipped ready by CMOSE and extracted by me from the raw video for DAiSEE and the private set), so the I3D “stream” is an MLP over that global motion/appearance summary rather than a temporal encoder; a per-window I3D sequence might add more and would let an I3D temporal encoder model real dynamics.
- (iii) **Small in-domain gain and a flat design space.** Most of the 243-configuration grid is not affected by which encoder it gets, and the best in-domain gain

over the baseline is small (+0.068 QWK). The decomposition is a sound, interpretable framework rather than a large jump in performance.

- (iv) **Weak DAiSEE signal.** The very low in-domain QWK on DAiSEE meant it could not be used for architecture development, so the cross-dataset conclusions rest mainly on CMOSE and the private set.
- (v) **Fixed training recipe.** A single seed, learning rate, and encoder size were used throughout; without multi-seed repeats the reported gaps are point estimates rather than confidence intervals, and small differences between nearby configurations should not be over-interpreted.

6.5 Practical Recommendations

For practitioners building engagement-monitoring systems, the study gives three concrete recommendations that do not depend on the specific architecture. First, select and optimise on ordinal agreement: make QWK the single metric that chooses models, report macro-accuracy and macro-MAE as secondary balanced checks, and treat plain accuracy as no more than a comparability number, because on these imbalanced ordinal labels accuracy can look healthy while the model has learned nothing useful about disengagement. Second, train on the combined datasets when the deployment data is unknown; combining datasets was the simplest tool that worked best for the unseen private set and needs no change to the model. Third, anticipate and monitor rare-class failure: the most important class, Highly-Disengage, is also the rarest and the hardest to detect, so a deployed system should treat low-confidence disengagement predictions with care and, where possible, leave the call to a human.

6.6 Suggestion for Future Works

- (i) **Targeting the rarest class.** Combine the architecture with fixes aimed at the imbalance (ordinal losses tuned for Highly-Disengage, minority oversampling such as SMOTE [15], or cost-sensitive decision thresholds) to recover HD recall without trading away Highly-Engage. Because HD is so rare, even small gains here would change the practical value of the system more than further QWK gains on the common classes.
- (ii) **Domain adaptation.** The cross-dataset gap calls for explicit unsupervised domain adaptation or domain-invariant representation learning, rather than the plain dataset combination that already helps. A small amount of labelled target data could be used to calibrate the model before deployment.

- (iii) **Learned, not listed, group encoders.** Replace the discrete 243-configuration search with a differentiable architecture search or a gating mechanism that picks per-group temporal operators end-to-end, so the model finds which encoder suits which group on its own instead of trying them all by hand.
- (iv) **Stronger and richer motion features.** Re-extract per-window I3D (or a modern video backbone) so the motion stream carries timing, and add audio/speech features, which CMOSE also provides, as extra feature groups within the same split-then-fuse framework.
- (v) **Frequency-domain analysis of engagement (exploratory).** A natural extra direction, not done in this thesis and left strictly as future work, is to study the temporal frequency of facial signals, under the idea that engaged behaviour is low-frequency and steady while disengaged behaviour is higher-frequency and restless, for example through short-time-Fourier-transform features or convolutions over the frequency axis.
- (vi) **Stronger evaluation.** Multi-annotator labelling of a larger private set, with reported agreement between annotators, and multi-seed training with confidence intervals would tighten every gap reported here and turn point estimates into statistically grounded claims.

6.7 Closing Remarks

Automatic engagement recognition is appealing exactly because online learning makes the webcam the only window onto the student, but the same setting makes the problem difficult: the labels are ordinal, severely imbalanced, subjective, and specific to each dataset. This thesis showed that keeping the structure of the facial features, by splitting them into behavioural groups and encoding each on its own terms, gives a sound, interpretable model that, combined with a motion stream and trained on combined datasets, generalises best to unseen, in-the-wild data. It also showed that the field’s usual reliance on accuracy hides a near-total failure to transfer across datasets, and that ordinal agreement metrics show it. Detecting the rarest, most important disengaged behaviour under distribution shift is still open, and remains the most important next step.

REFERENCE

- [1] C.-H. Wu **and** others, “CMOSE: Comprehensive Multi-Modality Online Student Engagement Dataset with High-Quality Labels,” *in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops* 2024, **pages** 4636–4645. DOI: 10.1109/CVPRW63382.2024.00466
- [2] A. Gupta, A. D’Cunha, K. Awasthi **and** V. Balasubramanian. “DAiSEE: Towards User Engagement Recognition in the Wild,” **urlseen** 26 **june** 2026. **url:** <https://arxiv.org/abs/1609.01885>
- [3] T. Baltrušaitis, A. Zadeh, Y. C. Lim **and** L.-P. Morency, “OpenFace 2.0: Facial Behavior Analysis Toolkit,” *in 13th IEEE International Conference on Automatic Face and Gesture Recognition (FG)* 2018, **pages** 59–66. DOI: 10.1109/FG.2018.00019
- [4] O. Çopur, M. Nakip, S. Scardapane **and** J. Slowack, “Engagement Detection with Multi-Task Training in E-Learning Environments,” *in Image Analysis and Processing – ICIAP 2022* **jourser** Lecture Notes in Computer Science, **volume** 13233, Springer, 2022, **pages** 411–422. DOI: 10.1007/978-3-031-06433-3_35
- [5] S. Malekshahi, J. M. Kheyridoost **and** O. Fatemi. “A General Model for Detecting Learner Engagement: Implementation and Evaluation,” **urlseen** 26 **june** 2026. **url:** <https://arxiv.org/abs/2405.04251>
- [6] F. M. Shiri, E. Ahmadi, M. Rezaee **and** T. Perumal, “Detection of Student Engagement in E-Learning Environments Using EfficientNetV2-L Together with RNN-Based Models,” *Journal on Artificial Intelligence*, **jourvol** 6, **number** 1, **pages** 85–103, 2024. DOI: 10.32604/jai.2024.048911
- [7] L. Zhang, X. Zheng, X. Chen **and** L. Cui, “Three-Dimensional View Relationship-Based Context-Aware Emotion Recognition,” *IEEE Transactions on Neural Networks and Learning Systems*, **jourvol** 36, **number** 7, **pages** 13 567–13 578, 2025. DOI: 10.1109/TNNLS.2024.3476249
- [8] I. Qarbal, N. Sael **and** S. Ouahabi, “Student’s Engagement Detection Based on Computer Vision: A Systematic Literature Review,” *IEEE Access*, **jourvol** 13, **pages** 140 519–140 545, 2025. DOI: 10.1109/ACCESS.2025.3596885
- [9] M. M. Santoni, T. Basaruddin **and** K. Junus, “Convolutional Neural Network Model based Students’ Engagement Detection in Imbalanced DAiSEE Dataset,” *International Journal of Advanced Computer Science*

- and Applications (IJACSA)*, **jourvol** 14, **number** 3, **pages** 617–626, 2023. DOI: 10.14569/IJACSA.2023.0140371
- [10] M. Tan **and** Q. V. Le, “EfficientNetV2: Smaller Models and Faster Training,” *in Proceedings of the 38th International Conference on Machine Learning (ICML)* **jourser** Proceedings of Machine Learning Research, **volume** 139, 2021, **pages** 10 096–10 106.
- [11] J. Carreira **and** A. Zisserman, “Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset,” *in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* 2017, **pages** 4724–4733. DOI: 10.1109/CVPR.2017.502
- [12] S. Bai, J. Z. Kolter **and** V. Koltun. “An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling,” **urlseen** 26 **june** 2026. **url**: <https://arxiv.org/abs/1803.01271>
- [13] S. Hochreiter **and** J. Schmidhuber, “Long Short-Term Memory,” *Neural Computation*, **jourvol** 9, **number** 8, **pages** 1735–1780, 1997. DOI: 10.1162/neco.1997.9.8.1735
- [14] A. Vaswani **and others**, “Attention Is All You Need,” *in Advances in Neural Information Processing Systems (NeurIPS)* 2017, **pages** 5998–6008.
- [15] N. V. Chawla, K. W. Bowyer, L. O. Hall **and** W. P. Kegelmeyer, “SMOTE: Synthetic Minority Over-sampling Technique,” *Journal of Artificial Intelligence Research*, **jourvol** 16, **pages** 321–357, 2002. DOI: 10.1613/jair.953
- [16] L. Hou, C.-P. Yu **and** D. Samaras, “Squared Earth Mover’s Distance-based Loss for Training Deep Neural Networks,” *arXiv preprint arXiv:1611.05916*, 2016.
- [17] D. P. Kingma **and** J. Ba, “Adam: A Method for Stochastic Optimization,” *in International Conference on Learning Representations (ICLR)* 2015.
- [18] J. Cohen, “A Coefficient of Agreement for Nominal Scales,” *Educational and Psychological Measurement*, **jourvol** 20, **number** 1, **pages** 37–46, 1960. DOI: 10.1177/001316446002000104
- [19] C. Spearman, “The Proof and Measurement of Association between Two Things,” *The American Journal of Psychology*, **jourvol** 15, **number** 1, **pages** 72–101, 1904. DOI: 10.2307/1412159
- [20] M. G. Kendall, “A New Measure of Rank Correlation,” *Biometrika*, **jourvol** 30, **number** 1–2, **pages** 81–93, 1938. DOI: 10.1093/biomet/30.1-2.81